DATA GOVERNANCE IN MOZAMBIQUE

Findings and Recommendations Based on a Rapid Diagnostic

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Acknowledgments

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Disclaimer

This diagnostic is not a comprehensive or representative assessment of the data ecosystem in Mozambique as a whole. It does not provide an overview of every kind of potential data initiative that could be developed. The diagnostic is not based on detailed, legal due diligence and does not constitute legal advice. Accordingly, this report should not be taken as a definitive statement on the topics it covers but should rather be used as an indicative guide for further evaluation of the enabling policy, legal, and regulatory framework (including institutional aspects) for data in Mozambique, and for further exploration of potential data use cases and opportunities.

This work is a product of the staff of the World Bank and African Development Bank with external contributions. The findings, interpretations, and conclusions expressed in this report do not necessarily reflect the views of the directors or executive directors of either institution or the governments they represent. Neither the World Bank Group nor African Development Bank guarantees the accuracy of the data included in this work.

List of abbreviations

AfDB	African Development Bank
ADE	National Geospatial Development Agency
AI	Artificial Intelligence
AMPETIC	Association of Information and Communication Technology Professionals & Companies
API	Application Programming Interface
AT	Tax Authority
BM	Bank of Mozambique
CGD	Citizen-generated data
CIUEM	Eduardo Mondlane University Informatics Center
CIUP	Pedagogical University of Maputo Informatics Center
CSO	Civil Society Organization
DNIC	National Directorate of Civil Identification (Ministry of Interior)
DNRN	National Directorate of Registries and Notary (Ministry of Justice)
DPA	Data Protection Authority
DPEC	Directorate of Planning, Statistics and Cooperation
EDGE	Digital Governance and Economy Project (World Bank)
ENPCT	National Company of Science and Technology Parks
IDEMOC	National Infrastructure for Spatial Data
INAGE	National Institute of Electronic Government
INCM	National Institute of Communications of Mozambique
INE	National Statistics Institute
INTIC	National Institute of Information and Communication Technologies
IT	Information Technology
MCTD	Ministry of Communications and Digital Transformation
MCTES	Ministry of Science, Technology and Higher Education (dissolved*)
MDAP	Mozambique Digital Acceleration Project
MEF	Ministry of Economy and Finance (dissolved*)
MIC	Ministry of Industry and Commerce (dissolved*)
MIMAIP	Ministry of the Sea, Waters Interiors and Fisheries (dissolved*)
MINEDH	Ministry of Education and Human Development (dissolved*)
MISA	Institute of Social Communication of Southern Africa
MISAU	Ministry of Health
MJCR	Ministry of Justice, Constitutional and Religious Affairs
MoF	Ministry of Finance
MTC	Ministry of Transport and Communications (dissolved*)
NSDI	National Spatial Data Infrastructure
ODRA	Open Data Readiness Assessment
OECD	Organization for Economic Cooperation and Development
SDG	Sustainable Development Goals
SPI	Statistical Performance Indicator
UEM	Eduardo Mondlane University
WDR	World Development Report

*Ministries tagged as dissolved were reorganized into newly created institutions by Presidential Decree in January 2025.

Executive summary

This report presents observations from a rapid diagnostic of the state of data governance and the data ecosystem in Mozambique. According to the World Bank, data governance "entails creating an environment of implementing norms, infrastructure policies and technical mechanisms, laws and regulations for data, related economic policies, and institutions that can effectively enable the safe, trustworthy use of public intent and private intent data to achieve development outcomes".[1]

The diagnostic aims to provide an overall assessment of the country's data governance framework, and to generate actionable recommendations for the Government of Mozambique to strengthen this framework in support of national development objectives. According to the government's Five-Year Development Plan (*Plano Quinquenal do Governo*), promoting good governance, efficient and transparent public administration, and digital delivery of public services to adopt a more diversified and competitive economy are of high priority.[2] Furthermore, the government has recently committed to elevating the digital agenda "to promote digital inclusion and boost the digital economy" in its 2025-2044 National Development Strategy[3] that is expected to be approved by the National Assembly by 2025. The diagnostic adopts a holistic approach, considering all stakeholders in the data ecosystem and their roles and priorities and focuses on the "soft infrastructure" of standards, policies, and institutions.

A robust data governance framework is critical in the digital age because it sets the parameters for data collection, management, and sharing within the national digital ecosystem, and supports the transition to digital service delivery. Data is the lifeblood of digital systems. As the Government of Mozambique is currently developing a national digital transformation strategy, it is an opportune time to take a comprehensive look at the country's data governance framework and ways to strengthen it in service of larger development objectives. The data governance framework would also underpin the government's investments in digital public infrastructure (DPI), including data exchange and digital IDs, and the digital public services built on top of it.

More broadly, effective data use can facilitate evidence-based policymaking; enhance public sector efficiency; empower entrepreneurs, innovators, and civil society; help address cross-cutting challenges such as sustainability and climate change; and over time support economic diversification and a broader digital transformation of society. A fit-for-purpose data governance framework is an essential enabler for these benefits of data to be realized. The data governance framework creates the conditions for government and non-government stakeholders to achieve more effective and equitable use of data, while safeguarding individuals, businesses, and societies from the risks to security and privacy that may arise.[4]

As is to be expected in a country with limited data and digital infrastructure and relatively low institutional capacity, there are barriers in Mozambique to fully harnessing the potential of data. According to the World Bank's Country Partnership Framework for Mozambique, weaknesses in the data environment "undermine the efficiency and efficacy of public resources in providing key public goods and erode confidence in the state".[5] Key challenges include the lack of a strategic approach to leveraging data for development goals, an inadequate policy and legal framework governing the use of data, and inadequate digital safeguards. Data management practices within the public sector are fragmented. Partly

[1] World Bank 2021a.
[2] Government of Mozambique 2020.
[3] IMF 2024.
[4] World Bank 2021a.
[5] World Bank 2023a.

as a result, the uptake of data for analytics, decision makin g, and innovation—both within and outside government—is relatively limited. Weaknesses in digital infrastructure and connectivity, and in the data skills base, are also important factors.

The Government of Mozambique has made progress in some key areas critical to fostering effective use of data and has committed to doing much more. In particular, the government has taken steps to strengthen the capabilities of its National Statistics Institute (INE) and of the National Geospatial Development Agency (ADE). INE now has substantially more technical capacity to produce high-quality data that can be used for evidence-based policy making, and ADE has significantly increased the government's ability to leverage geospatial data. The government is preparing several strategies with relevance to the data agenda, including a national digital transformation strategy, a public sector digital transformation strategy, an updated national statistical strategy, and an artificial intelligence (AI) strategy.[6] Through the World Bank's Mozambique Digital Acceleration Project (MDAP) and Digital Governance and Economy (EDGE) Project, the government has committed to building the foundations of digital transformation and expanding the government's capacity to produce, store, share and use data safely. This includes strengthening the institutional and policy framework for data privacy and protection, identity management, open data and interoperability.

This diagnostic was undertaken in partnership with the Government of Mozambique by a joint World Bank and African Development Bank Group (AfDB) team. It was initiated at the request of the Instituto Nacional de Tecnologias de Informação e Comunicação (INTIC), Mozambique's ICT regulator, an institution supervised by the Minister of Communications and Digital Transformation (MCTD). The initiative aims to help the government set the stage for a vibrant data ecosystem in which civil servants can leverage data for decision-making, service delivery, and efficient public administration, and the private sector and civil society can harness the power of data to create jobs, grow the economy, and help build a more inclusive society. Government agencies that are involved in data production, management, and/or use, and their partners, are the key audience for this report. The diagnostic aligns with the World Bank's Country Partnership Framework, emphasizing greener, more resilient, and inclusive development, the AfDB's Data Innovation Lab initiative, and the government's digital agenda as laid out in the updated National Development Strategy.

The methodology involved a review of existing literature, World Bank documentation, and other secondary sources, plus consultations with government and non-government stakeholders. Following a preliminary desk review and stakeholder mapping, the team conducted a series of interviews and consultations in close partnership with government counterparts during the rapid data diagnostic mission. The World Bank's Open Data Readiness Assessment (ODRA) tool provided an organizing framework for the consultations, but it was expanded and adapted to align with the focus on data governance.

Building a robust data governance framework is a process that takes time and investment, but there are concrete steps the government can take now—some of which are already underway—to create a stronger enabling environment for data use in a relatively short period. The recommendations focus on a set of short- to medium-term actions, taking a maturity model approach. In the longer term, the government can continue to bolster the policy and legal underpinnings for data use along with the capacity both within and outside government, especially the private sector, to extract maximum value from data, including in economic sectors.

[6] Under the World Bank's EDGE project the Government of Mozambique is preparing an AI strategy. Preparation of an AI law is not currently a priority given more pressing gaps in other aspects of the legal framework for data governance.

One guiding principle of this report's approach is to mainstream attention to data governance into existing institutions and strategies that are already under development, rather than creating new structures or stand-alone policies that would increase fragmentation. The government is currently working to develop several strategies where data is a key component, as mentioned above. Moreover, it has already created institutions for designing and implementing aspects of its data governance framework. This report proposes actions that could be taken within these established initiatives to maximize efficiency and sustainability. At the same time, inter-ministerial coordination will be an essential ingredient to harmonize practices and approaches throughout the government and across the data lifecycle.

The diagnostic's recommendations focus on developing a strategic and legal framework for data governance, improving public sector data management, strengthening institutional capacities and the data skills base, and understanding the needs of data users both inside and outside government. The report presents recommendations grouped under five themes that progress from foundational and structural concerns to operation and implementation as follows, with details in Table A:[7]

1. **National data strategy and senior leadership.** In the absence of a unified national vision on the role of data in national development, government entities pursue their own data-related initiatives and create and follow their own policies and procedures. The report recommends developing a strategic approach to data that would articulate the foundational role of data in national development, unify fragmented and overlapping efforts, and more fully make use of the government's existing data assets.

2. **Institutional arrangements.** To ensure a consistent approach to data governance and management across government, and to reduce duplication of effort, it is important to clarify institutional responsibilities. Key suggestions are to ensure a collaborative approach while clearly designating certain entities as responsible for key roles, particularly with respect to data stewardship and standards, which will be important in the short- to medium-term as the government seeks to deepen its use of data and implement an interoperability framework.

3. **Policy and legal framework for data governance.** There are some gaps in the policy and legal framework for data in Mozambique. The report offers recommendations related to the government's preparation of a draft Data Protection Law and laying the groundwork for developing wider-ranging data policies covering the entire data lifecycle.

4. **Procedures, practices, and standards for data management.** By defining data standards and procedures in the formal data governance framework, and preparing a Data Classification Policy, the government can better ensure data quality, interoperability, accessibility, and security, facilitating decision making and more efficient public service delivery.

5. **Capacity building.** Data skills—both technical and non-technical—are increasingly necessary to function in the workforce and society. Apart from the technical skills required to manage data through its lifecycle, advanced data governance literacy is essential for digital business process management and to develop digital-first operational models. Data skills are similarly vital for citizens, consumers and regulators to navigate the complex impact of data on market

[7] The report focuses on the data governance framework as created by the government in the context of its interactions with the civil society and the private sector. The data-driven transformation of the private sector is outside the scope of the report but is a major opportunity area for Mozambique.

structures, and consumer and citizen rights. It is also critical to develop practices to attract and retain digital skills in a highly competitive market for talent.

Table A. Recommendations to consider by theme

Theme	**Specific Recommendations**
1. National data strategy and senior leadership	• Establish an Inter-Ministerial Digital Transformation Steering Committee to strengthen coordination. • Through a multistakeholder, consultative process led by the Steering Committee, establish a shared vision for the role of data in the digital ecosystem and articulate its fundamental role in achieving national objectives. • Ensure all strategies that play a role in advancing the data agenda (including on artificial intelligence [AI]) are aligned with and refer to the overarching framework enshrined in the national digital transformation strategy. • Establish mechanisms for regular engagement with data users, including government analysts, the private sector, academia, the media, and civil society, to surface challenges and priorities within the data agenda. • As part of a demand-driven strategy for investing in data, identify high-value data use cases to demonstrate the value proposition for data in Mozambique.
2. Institutional arrangements	• Ensure that key data institutions (INTIC, INAGE, INE, INCM, ADE, the BAÚ - Single Service Desks[8] and others) are included in the proposed Inter-Ministerial Digital Transformation Steering Committee and the working groups it oversees. • Clarify institutional roles and responsibilities for government entities. i. Host of government data catalog and data standards. ii. Data steward. iii. Data leaders at the sectoral level. • Capacity building and adequate funding will be necessary to enable existing institutions to take on these additional responsibilities.
3. Policy and legal framework for data governance	• Adopt a data protection law aligned with best international practices, while being sensitive to the local context. • Create and operationalize the National Data Protection Authority, giving it adequate institutional mandate and resources to be sustainable and independent. • Revisit provisions of the Licensing Regulation[9] in light of broader data governance objectives. • Strengthen implementation/enforcement of existing laws and lay the groundwork for developing wider-ranging data policies covering the entire data lifecycle, including aspects and concerns related to the use of data by AI systems and platforms.

[8] The Single Service Desks (Balcão de Atendimento Único) are one-stop-shops that work to streamline procedures for the licensing of economic activity and other related public services, including via the e-BAÚ online platform. The BAÚ are supervised by the Ministry of Industry and Commerce (MIC).
[9] Regulation for the Registration and Licensing of Intermediary Providers of Electronic Services and Digital Platform Operators

Theme	Specific Recommendations
4. Procedures, practices, and standards for data management	• Design and implement a Data Classification Policy defining the processes and standards that comprise a national security classification system, and prepare the respective regulations. • Conduct an inventory of the data that currently exists within government to identify areas of complementarity, duplication, and overlap, and need, as well as to classify data by level of security to ensure appropriate data storage. • Determine an institutional owner for each category of data to ensure there is a "single source of truth". • As an indicative example of an initial pilot of a multi-stakeholder approach to interoperability, the government could consider establishing a Geographical Names Register based on the work of ADE and IDEMOC. • Building on the proposed Geographical Names Register pilot, INAGE can spearhead development of a common data exchange infrastructure (potentially leveraging the DPI model), standard practices on data storage and sharing, and central data registers to enable interoperability and data sharing. • Create a mechanism to improve coordination among key data stakeholders within government in terms of data practices, infrastructure, and technology. • Support expanded and more streamlined public access to data, taking an approach informed by what current and potential data users need. • Develop guidelines for measuring the impact and quality of data-driven services (i.e., digital public services), and publish the data online to increase transparency and accountability. • Develop and implement standardized controls and procedures to ensure data protection and security across government agencies: i. Enhanced data security measures. ii. Detailed documentation and record-keeping. iii. Detailed data breach notification requirements. iv. Comprehensive compliance and audit processes.
5. Capacity building	• Conduct an inventory of data skills gaps in key roles within government institutions that are important to the data governance framework. • Think beyond technology and focus on building competencies such as digital business process engineering, digital business and operational models, digital market structures, and digital consumer and citizen rights. • Establish general data training programs for the civil service at different career stages and invest in continuous upskilling. • Establish specific training courses on aspects of data governance for select staff of key agencies who need to be well versed. • Talent management: fellowships/rotations for data (e.g., two years in different agencies). • Cross-pollinate good practices across the government and encourage learning by doing.

The report is structured as follows. The first section provides background on the diagnostic and the state of the data ecosystem in Mozambique, followed by context on leveraging data effectively for development drawn from the international literature. The second section summarizes the main findings, organized by the pillars of the ODRA, and the third section suggests recommendations for the government to consider. The report concludes with a brief section on next steps, followed by Appendices that include a full list of stakeholders consulted for the report, more detail on the methodology, and other background information.

1. Introduction

Background and methodology: The Mozambique data governance diagnostic

In recent years, Mozambique has faced domestic and external shocks that have dampened growth and caused development setbacks, but the economic outlook is now more promising, and the country's natural resource wealth suggests the potential for an inclusive, resilient future. The government has made significant progress in areas including economic management and disaster risk management in the wake of a debt crisis, a pair of cyclones that caused substantial damage in 2019, and the COVID-19 pandemic. However, like many other low-income countries, Mozambique is severely off-track on meeting the SDGs, ranking 149th out of 166 on the Sustainable Development Goals (SDG) Index Rank and with around a quarter of the targets worsening rather than improving.[10] Mozambique is also one of the world's most vulnerable countries to climate change, ranking 173rd out of 192 countries on climate readiness.[11]

Mozambique has relatively low levels of connectivity and digital infrastructure, as well as low levels of digital and data literacy, making it difficult to leverage the power of data and digital technology to tackle development challenges. Less than a fifth of the population has access to the Internet at home and only three percent of households have a computer.[12] Most government agencies and services have not been digitalized yet. The low level of digital skills among the population is also a challenge, with 14 percent of those who are offline saying that they do not know how to use the Internet.[13]

In terms of public sector digital transformation, Mozambique falls in category "C" in the World Bank's GovTech Maturity Index (GTMI), indicating "some focus" on digital transformation of government, but still the potential of digital technology is largely untapped.[14] Digital public services that do exist often include steps that require manual processes or paper-based documentation, and there is limited or no data sharing across applications—users must enter the same information in multiple websites. Poor administrative and public financial data further hampers data-driven decision-making. For example, in Mozambique's health sector, there is a lack of detailed expenditure classification at the subnational level that stems from weak capacity for data entry in health facilities. This results in a lack of data for effective resource planning and monitoring.[15]

Key real economy sectors such as natural resources, health, and agriculture are in nascent stages of digitalization. Few systems exist to capture and collect data systematically and at scale, making it difficult to measure and increase the effectiveness of public spending. Looking at another sector, there is a need for "open access platforms for sharing hydrological data and information"[16] to confront the problem of water scarcity in the country.

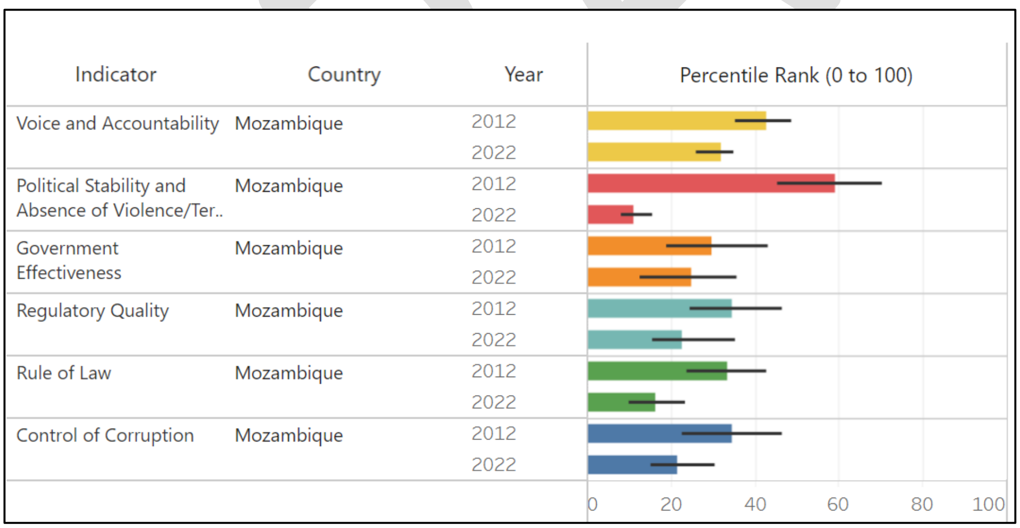
Leveraging data more effectively is essential to increasing Mozambique's resilience and adaptive capacity in the face of climate change by enabling risk modeling, disaster response, water and forest resource management, and more. According to the World Bank's *Country Climate and Development Report* for Mozambique, it is a priority for the country to boost institutional capacity around data

[10] https://dashboards.sdgindex.org/profiles/mozambique
[11] https://gain.nd.edu/our-work/country-index/rankings/
[12] ITU data, available at https://datahub.itu.int/data.
[13] World Bank 2022a.
[14] https://www.worldbank.org/en/programs/govtech/gtmi
[15] World Bank 2021b.
[16] World Bank 2023b.

availability and quality, "not least to position the country to mobilize additional climate funding sources that require sound monitoring, reporting and verification (MRV) of greenhouse gas emissions."[17] The government's *Disaster Risk Management Master Plan 2017—2030* highlighted the importance of data, as it suggested "the development of systems for collecting and managing data on the occurrence and impacts of disasters, as well as initiatives to insert risk reduction and resilience-building criteria into the processes of planning at all levels of government".[18] Expansion of data-driven systems for digital cash transfers would improve disaster response, and digital systems more generally would help enable continuity of public services in times of crisis. Excellent hyperlocal data is also key for meeting other environmental goals, for example Mozambique's commitment under the Bonn Challenge to restore one million hectares of degraded land by 2030.[19]

Investing in data can help government become more responsive and effective in meeting the needs of its citizens. In light of Mozambique's recent history and the challenges it faces, the World Bank's Systematic Country Diagnostic proposes that there is a need for a "greater focus on governance and accountability… and a stronger social contract".[20] In the most recent Afrobarometer survey, 41 percent of respondents felt that the country is going in the wrong direction (a sentiment that was more prevalent in urban areas, where 51 percent of people said that it is going in the wrong direction, versus 36 percent in rural areas).[21] As shown in Figure 1, Mozambique's percentile rankings in the Worldwide Governance Indicators have declined over the past decade. Part of a strategic approach to data for Mozambique could be to consider how better information—in the hands of policymakers, business, and citizens—could help to address these challenges.

Figure 1. Change in Worldwide Governance Indicators for Mozambique from 2012 to 2022



Source: https://www.worldbank.org/en/publication/worldwide-governance-indicators/interactive-data-access

The World Bank Statistical Performance Indicators (SPIs) offer some insight into how Mozambique is doing in terms of "how well, how broadly, and how frequently national statistical systems

[17] World Bank 2023b.
[18] World Bank 2023b.
[19] World Bank 2023b.
[20] World Bank 2021b.
[21] Ipsos 2022.

collect, produce, and disseminate high-quality data in a publicly accessible manner".[22] Mozambique's overall score (59 out of 100) is on par with the average for Sub-Saharan Africa and slightly higher than the average for low-income countries (55 out of 100).[23] Looking at Table 1, which shows Mozambique and select comparator countries and their overall SPI scores along with the pillars[24] that comprise it, it seems that the weakest areas for Mozambique relate to its relatively narrow universe of data sources, and gaps in its data infrastructure (as defined by the SPI).

Table 1. Mozambique and Comparator Countries' Scores on the Statistical Performance Indicators (SPIs)

	SPI Overall Score	Pillar 1: Data Use	Pillar 2: Data Services	Pillar 3: Data Products	Pillar 4: Data Sources	Pillar 5: Data Infrastructure
Mozambique	**58.7**	**70**	**60**	**79**	**37**	**55**
Botswana	61.2	50	69	78	64	45
Ethiopia	61.1	90	65	79	33	40
Kenya	66.3	90	60	76	35	70
Madagascar	53.7	60	61	78	25	45
Malawi	64.8	90	62	80	46	45
Nigeria	58.6	80	64	73	31	40
Rwanda	70.6	90	71	76	53	60
South Africa	82.4	80	86	88	73	85
Uganda	70.7	100	65	80	40	70
Zimbabwe	70.2	100	67	86	38	60
	Key:	**80-100**	**60-79**	**40-59**	**20-39**	**0-19**

Data source: https://www.worldbank.org/en/programs/statistical-performance-indicators

Reflecting the low diffusion of data-driven digital services and applications, use of data within business and society in Mozambique is relatively limited, but to some extent the private sector has acted on the commercial potential of data-driven products and services. Digital finance is currently among the most digitally sophisticated sectors in Mozambique with services such as Vodacom's M-Pesa and Movitel's e-Mola serving millions of domestic customers. E-commerce companies such as Xava and Compra have demonstrated potential in the marketplace. These developments suggest that there is likely latent demand for government leadership on improving the ecosystem for data-related products and services.

A stronger data governance framework could enable all stakeholders—from government to the private sector, media, academia, citizens, and others—to use data effectively and safely to advance national development objectives, and to accelerate digital transformation. The OECD notes that data governance "ranks as a top priority for governments aiming to maximize the benefits of data while addressing challenges such as privacy and intellectual property as well as competition and empowerment"[25]. The lack of a robust data governance framework means that Mozambique is missing a key enabler of public sector transformation, government accountability, an attractive investment climate, climate resilience, and the transition to a digital economy in the age of artificial intelligence (AI) and other emerging technologies.

[22] https://www.worldbank.org/en/programs/statistical-performance-indicators
[23] https://datanalytics.worldbank.org/SPI/?tab=country-reports
[24] See Appendix 3 for definitions of the pillars.
[25] https://www.oecd.org/en/topics/sub-issues/data-governance.html

The Mozambique Data Diagnostic was initiated by the African Development Bank Group (AfDB) and the World Bank in response to a request from the Government of Mozambique's National Institute of Information and Communication Technology (INTIC), the national ICT regulator, an institution supervised by the Minister of Communications and Digital Transformation (MCTD)[26]. INTIC proposed a diagnostic exercise that would help the government to identify opportunities and gaps in the policy, legal, technical, and institutional foundations of data governance in the country. According to *World Development Report 2021: Data for Better Lives,* data governance "entails creating an environment of implementing norms, infrastructure policies and technical mechanisms, laws and regulations for data, related economic policies, and institutions that can effectively enable the safe, trustworthy use of public intent and private intent data to achieve development outcomes".[27] The diagnostic goes somewhat beyond this definition however to adopt a "data ecosystem" approach, meaning it takes a holistic look at the context in which data is produced and used. This includes "supply" side issues like the policy and legal framework, government data collection and production, and government data management practices across the data lifecycle (national digital infrastructure is also an essential prerequisite but out of scope for this work). To a lesser extent it also includes "demand" side issues like uptake of data for analytics within government, and the needs and role of non-government stakeholders in the data ecosystem, which include the private sector, civil society, academia, the media, international partners, and others.

The diagnostic is intended to provide the Government of Mozambique with actionable recommendations in support of its ambition to leverage data for development, and to inform implementation of two World Bank projects currently underway. The diagnostic has generated high-level recommendations and specific action steps grounded in a whole-of-government approach to data governance, aimed at enabling effective and responsible use of data in the service of national development objectives. Specifically, the Data Diagnostic will help inform work under the World Bank's Mozambique Digital Acceleration Project (MDAP) and Digital Governance and Economy (EDGE) project, which includes supporting the adoption and implementation of a national digital transformation strategy, a public sector digital transformation strategy, an interoperability framework, and a data protection law, among other key elements.[28] The diagnostic also aligns with the AfDB's ongoing efforts to help Mozambique establish a system to improve data accessibility to support accountability and facilitate monitoring national programs, the Sustainable Development Goals (SDGs), and the African Union's Agenda 2063 in the country.

The methodology for the diagnostic involved secondary research and semi-structured stakeholder consultations[29] organized around the eight pillars of the Open Data Readiness Assessment (ODRA) tool[30] developed by the World Bank[31]. To better align with the government's priorities, ongoing World Bank operations and the AfDB's Data Innovation Lab initiative, the team adapted the ODRA tool by expanding its scope beyond open data to encompass the role of all types of data, data interoperability, and data

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[26] At the time the diagnostic was launched, INTIC was supervised by the Minister of Science and Technology, Higher and Technical and Vocational Education (MCTES). That ministry was dissolved, and its technology area was absorbed into the newly created Ministry of Communications and Digital Transformation by a Presidential Decree in January 2025.

[27] World Bank 2021a.

[28] This report focuses on the "soft" elements of data governance, as requested by the government. Digital infrastructure and cybersecurity are outside the scope of the current diagnostic.

[29] Please see Appendix 1 for full list of stakeholders consulted during the mission and virtual interviews.

[30] The team adapted ODRA Version 3.1, prepared by the World Bank's Open Government Data Working Group; available at http://opendatatoolkit.worldbank.org/en/data/opendatatoolkit/odra.html.

[31] The World Bank has deployed the ODRA in about 30 countries over the last decade to help governments assess their readiness to establish, manage, and benefit from open data programs.

governance in the public sector more broadly. The diagnostic focuses on the most relevant aspects of the ODRA, including leadership and strategic vision, policy and legal framework, institutional setup, technical capacity and data-related practices. Technical aspects including physical infrastructure and cybersecurity, while key enablers of the data agenda, are largely outside the scope of this diagnostic and therefore not addressed in the recommendations. The diagnostic included the private sector in the consultation process, but the recommendations are aimed at the government.

The diagnostic approach is informed by other work undertaken by the World Bank and AfDB. In particular, it relies on *World Development Report 2021: Data for Better Lives* and the framework it proposes for a new social contract for data. This social contract would enable government and non-government stakeholders to extract maximum benefits from the use and reuse of data, while protecting against the harms that can arise from data misuse. The diagnostic also benefited from inputs taken from the questionnaire used by the AfDB for the assessment of the degree of preparedness of the African data ecosystem in developing the Data Innovation Lab.

It should be noted that this diagnostic is not a formal or comprehensive evaluation and does not provide specific legal advice. The diagnostic is intended to serve as a rapid assessment to help the government identify areas that warrant attention and prioritize reforms and new initiatives accordingly. The report is written from a policy perspective and given time and resource constraints does not include in-depth analysis of every component, with prioritization driven by client needs, maturity of the data ecosystem, and areas of opportunity for reform or investment.

Context: Data and Development

Effective use of data in the public sector can be an important enabler of economic and social development. Timely, reliable, relevant data allows officials and civil servants to design appropriate policies, monitor the results of government programs, and calibrate their interventions based on the evidence.[32] Data also points the way toward a more inclusive, equitable society, as it allows for a better understanding of who is being served and who is left out of current programs and systems.[33] Without data, governments are "driving blind" on their development journey. Data analytics can enable governments to target resources more precisely so that they will have the greatest impact. For example, Kenya's National Police found that most accidents happened on only 150km of the country's 6,200km of roads, and were then able to allocate resources to the places that mattered.[34] Moreover, when citizens and non-government stakeholders have access to good data and the ability to use it, they can hold governments to account for the outcomes of public policies and investments.

Data is critical for government and civil society to monitor development goals and respond to crises. From the Sustainable Development Goals (SDGs) to the Paris Agreement's climate targets and the African Union's Agenda 2063, countries have set a range of targets that are meant to be monitored via data "unprecedented in their scope and granularity".[35] Moreover, the collection and sharing of data can be essential in a crisis, as in the case of the 2014-2015 Ebola outbreak in West Africa, when analysis of public and shared data was essential to confronting the emergency.[36] Sometimes the value of data is most striking in its absence, as in the case of a lack of data on climate in Africa, which has impeded the design

[32] World Bank 2021a; van Ooijen, Ubaldi, and Welby 2019.
[33] Joliffe, Mahler, and Veerappan, et al. 2021.
[34] Mo Ibrahim Foundation 2024.
[35] Mo Ibrahim Foundation 2024.
[36] United Nations 2020.

of effective policies to fight climate change.[37] Ndemo et al. (2023) suggest that in many countries in Africa, "critical socioeconomic data are collected in unreasonable time intervals, are rarely available to inform policy on time, and are not accessible to researchers to conduct studies to build the stock of knowledge that could be transformative".[38]

In the digital age, a vibrant national data ecosystem can be a key driver of private sector development. Efficient, data-driven public services strengthen a country's investment climate and enable growth and innovation by making it easier to start a business, conduct international transactions, comply with laws and regulations, pay taxes, clear customs, secure property rights, and the list goes on. Given sufficient data infrastructure and skills availability in the economy, the private sector can harness the power of data and digital services to deliver better products and thus contribute to job creation and economic growth. The AfDB has estimated that the economic potential in Africa of open data alone could be equivalent to 1-2 percent of regional GDP.[39] Analysis commissioned by the Global Partnership for Sustainable Development Data estimated an "average return of US$32 for every dollar invested in data systems, spread across sectors".[40]

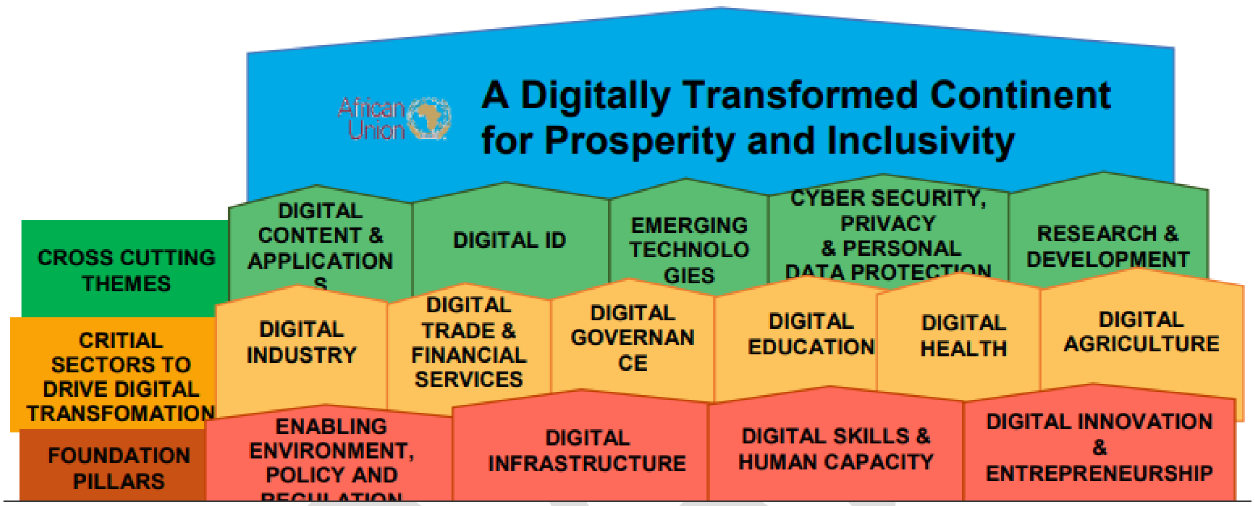
At the same time, the potential harms from data's misuse are significant, and the benefits of data described above are difficult to realize without an adequate data governance framework in place. Potential harms that can arise include threats to data privacy and security, as well as risks of data "misuse", i.e. that data will be used inappropriately by governments for surveillance, by individuals and organizations to enable criminal activity, or by firms to unfairly reinforce marketplace power.[41] As noted by the Global Partnership for Sustainable Development Data's latest strategy, as technology advances "the potential for data to reinforce structural inequalities is greater than ever, and attention to the power relationships underpinning data systems is more critical".[42] The data landscape has been changing so quickly in the digital age, with a proliferation of data itself and of data sources, that governments have struggled to keep policy and legal frameworks adequate to the task, resulting in a data "policy gap".[43] A robust data governance framework builds investor and citizen confidence and trust by ensuring data protection and security and guarding against the misuse of data, so that the manifold benefits of data for development can be realized.

At the national level, the data agenda and the digital agenda are closely intertwined. For societies to realize the transformative potential of transitioning to a digitally enabled economy, and to implement digital service delivery, they must pay attention to how data is governed and managed. One of the broadly-endorsed "Principles for Digital Development" is to establish "People-first data practices [that] prioritize transparency, consent, and redressal while allowing people and communities to retain control of and derive value from their own data".[44] The UN's nascent Global Digital Compact[45] has named data governance as one of the four pillars of the Compact, creating an opportunity to define a baseline for data governance worldwide.[46] The African Union's "Digital Transformation Strategy for Africa (2020-2030)" includes cybersecurity, privacy and personal data protection as a cross-cutting theme (Figure 2). And just

[37] UNFCCC 2020.
[38] Ndemo, Ndung'u, Odhiambo, and Shimeles 2023.
[39] AfDB 2017.
[40] https://www.data4sdgs.org/initiatives/power-of-data-unlocking-data-dividend-sdgs
[41] World Bank 2021a.
[42] Global Partnership for Sustainable Development Data 2023.
[43] PARIS21 and Mo Ibrahim Foundation 2021.
[44] https://digitalprinciples.org/. The principles have been endorsed by more than 300 development organizations.
[45] https://www.un.org/techenvoy/global-digital-compact
[46] Slotin and Mclaren 2024.

as data governance and management are essential to the success of digital transformation, so is digital development critical to governments' ability to reap the benefits of data. For these reasons, governments need to consider their digital and data ambitions in an integrated way, as in the case of Rwanda (Box 1). According to the World Bank's GovTech Maturity Index, 176 economies (about 90 percent) already have or are about to adopt a GovTech or digital transformation strategy.[47]

Figure 2: African Union: The Digital Transformation Strategy for Africa (2020-2030)



Source: African Union 2020.

Box 1. Rwanda's path to data-driven transformation

Rwanda's approach to data-led digital transformation illustrates a combination of high-level political support, a strong institutional setup, a robust data governance framework, technological innovation, and continuous capacity building. Recognizing the critical role of accurate data in an intensive knowledge and data-driven economy as articulated in its Vision 2050[48], the Rwandan government developed and adhered to a national statistical plan aligned with both national and regional development goals. This approach enabled strategic allocation of resources and foreign aid, advancing the country's priorities. Additionally, the government empowered the National Statistical Institute of Rwanda (NISR) to evolve beyond its traditional role to serve as a "data steward" with a broader view of data types and sources, providing the necessary support and resources to ensure its success. The investment in NISR facilitated the adoption of technological innovations[49] that played a pivotal role in its achievements. The Institute implemented modern sampling techniques, leveraged geo-spatial data, and utilized Computer-Assisted Personal Interviews (CAPI), significantly enhancing data collection efficiency and accuracy. These technological advancements, coupled with rigorous training programs for staff, enabled NISR to produce high-quality data that informs policy and drives economic growth. Additionally, NISR surveys all users of official statistics[50] from government agencies to private sector users biennially to ensure NISR meets users'

[47] World Bank 2022b.
[48] Vision 2050.
[49] Jütting 2023.
[50] Ndemo, Ndung'u, Odhiambo, and Shimeles 2023.

priority requirements. Its training center launched in 2019 is designed to lay groundwork to make Rwanda a big data regional hub for Africa.[51]

In a broader context, Rwanda's commitment to data supports its overarching digital transformation strategy. It has made strides on its digital agenda[52] by enhancing its data governance framework including enacting a law on Data Privacy and Protection (DPP), stipulating mechanisms for coordinating production and dissemination of statistical data, ensuring safe electronic transactions, hosting critical government information in one central national data center, launching initiatives like the Irembe platform (citizens' one-stop-shop for 88 cross-sectoral e-government services), establishing strategic partnerships with international actors (e.g., Mastercard and the Gates Foundation), and using data for guiding policy. Rwanda also recently unveiled a national AI policy to create and advance its AI ecosystem and position itself as a responsible AI champion.

2. Main findings

Several themes, opportunities, and challenges emerged from the research and consultations undertaken for this data diagnostic. Overall, Mozambique's national capacity to leverage data is constrained by gaps in leadership, coordination, data management, and available technical and human resources. Key issues include a lack of centralized guidance on data protection and security protocols, policies and formats for public data release, data sharing and interoperability between government departments, data management and storage, data quality and adherence to international standards, and operational channels for joint public and private data initiatives (reflected in a lack of public/private collaboration schemes regarding data acquisition and use). In addition, limited staff and technical capacity for data analytics hinders the government's ability to use data effectively.

Weaknesses in data governance and management and the data skills base affect more than the government's use of data. These limitations also undermine the ability of the private sector and civil society to use data for innovation, business growth, and initiatives to improve citizens' well-being. As AI and other new digital technologies grow in economic importance, a holistic approach to data governance and related investments will lay a strong foundation for the uptake of these technologies. Opportunities and programs in Mozambique for upskilling in data science and related areas are few for civil servants, the private sector, and students alike. Non-government stakeholders find it difficult to access and use government data due to the lack of data availability in digital formats. Government can be slow to respond to requests for data, and at the same time, public agencies lack user feedback on their data.

The government has made progress in some important areas and is working on several data-related strategies and investments. So far, the government's primary areas of focus on the data agenda have been on strengthening official statistics, enabling digital transformation of service delivery, and increasing data security and protection, and these efforts are bearing fruit. There has been relatively less attention to fostering data sharing, reuse, and analytics, including by harnessing the value of administrative data generated by line ministries. In general, investment is needed in strategic and regulatory frameworks, data management guidelines, and digital and data literacy. Addressing these issues requires high-level leadership and cross-sectoral collaboration to create an inclusive data ecosystem that supports sustainable development and innovation across the country. Table 2 summarizes the main findings of the diagnostic, which are then discussed in more detail. Findings are organized by the eight pillars of the ODRA

[51] https://unstats.un.org/bigdata/regional-hubs/rwanda-concept-note.pdf
[52] Ingabire 2024.

methodology (slightly adapted, as explained in Appendix 2). Please note that this is not meant to be a comprehensive assessment and the findings below must be considered provisional.

Table 2. Main Diagnostic Findings by Pillar[53]

Pillar	Importance	Commentary
Senior Leadership: *Vision set and implemented by senior leadership on data-led digital transformation*	Very high	Although Mozambique's Information Society Policy lays out a vision for an ICT-enabled public sector and society, the data governance agenda has not benefitted from the high-level leadership and visibility required to drive reforms to support the economic diversification and transformation emphasized in the Five-Year Development Plan for 2020–2024 (*Programa Quinquenal do Governo* - PQG).
Policy/Legal Framework: *Comprehensiveness of the framework for data-led digital transformation*	Very high	The government has adopted several pieces of the policy/legal framework for data governance, but some significant gaps remain. There are laws that touch upon data and privacy obligations, but dedicated legislation on data protection, data privacy, and other safeguards has not been approved. The government is currently preparing a proposed Data Protection Law. Right to Information legislation has been around for a decade, though it could be more comprehensive and more consistently enforced. Other relevant legislation includes the Electronic Transactions Act, the Microdata Dissemination and Access Policy, and more.
Institutional Structures, Responsibilities and Capabilities within Government: *Institutional responsibilities and capacities for data management and coordination*	Very High	While INTIC under the leadership of MCTD has a mandate for setting the national data vision and strategy, its limited capacity (in terms of financing and number of staff) constrains its ability to implement the mandate, particularly with respect to monitoring compliance. There are also other government institutions (INE, ADE, INCM) that have related mandates within the data ecosystem. Although Mozambique has some institutions with good capabilities relevant to the data agenda, the institutional landscape is complex and fragmented, with little coordination across ministries and agencies. The lack of coordination creates a blurring and dilution of responsibilities.
Government Data Management Policies, Procedures and Data Availability: *Including standards and norms*	High	Generally, government agencies have developed their own internal policies and procedures around data production and management. Some do follow international good practices, but the fragmentation of data systems, classifications and approaches undermines the government's ability to leverage data assets for analysis and decision making, leads to inefficiencies and duplication of effort, and heightens data security and privacy risks. Interconnectivity is weak.
Demand for Data and Data Services + Civic Engagement: *Stakeholders' access to and use of data*	High	The government's 2022 Microdata Dissemination and Access Act envisions engaging with users to understand their requirements, indicating an increasing focus on ensuring that public data is relevant and useful. The Information Society Policy includes a commitment to enhancing citizen participation through e-government initiatives. However, currently the process for citizens to request government data can be tedious, with inconsistent levels of responsiveness from government entities, and user engagement and feedback mechanisms have yet to be established.
Funding a Data Program: *Funding sources and staffing allocations for data programs*	Very high	Agencies consulted for the diagnostic indicated a lack of dedicated and reliable funding from the government to sustain data initiatives, as well as host and maintain information systems, after a certain start-up phase.

[53] The pillars are defined by the Open Data Readiness Assessment toolkit, with minor modifications. See Appendix 2 for details.

Pillar	Importance	Commentary
Technology and Skills Infrastructure: *Digital infrastructure, connectivity and access; data and digital skills and education*	Very High	Technology and skills infrastructure within the government vary widely by agency, and in the country as a whole, connectivity and the data/digital skills base are relatively weak. These are key determinants of Mozambique's ability to collect, analyze, and share data effectively. Within the government, a lack of dedicated formal training programs to upskill civil servants in data and advanced analytics topics means that civil servants have few opportunities to expand their skills, and there are several challenges that make it difficult to recruit and hire new staff. While some agencies have partnered with international players and academia to develop training for their technical staff, others have not been able to do so. More broadly, there is a need to develop data analytics and data science academic programs and curricula to prepare the country's workforce for an age of digital transformation.
Artificial Intelligence and Machine Learning: *Readiness and adoption*	Low	Use of artificial intelligence and machine learning technologies is at a nascent stage in Mozambique, though there is interest from some government entities in finding ways to leverage it, and the government is currently preparing an AI strategy.

A. Senior Leadership

The government's ability to leverage data effectively requires high-level leadership, a problem-driven approach, and commitment to the data-led transformation agenda. The Government of Mozambique has yet to articulate a national vision around the potential of data in the digital age to strengthen policy making and government effectiveness, boost economic growth and job creation, and enable inclusive, sustainable development. Partly as a result of a lack of top-down messaging around the value and importance of data from the highest level of government, data usage and exchange within the Mozambican government is currently limited, and there is minimal engagement with non-government stakeholders around the data needs in the broader data ecosystem.

Mozambique's Information Society Policy (2018) is a 10-year plan that outlines the government's ambition to integrate ICTs across various sectors, promote inclusive growth, and set the stage for a digitally-driven society. The Information Technology Policy (ICT Policy) passed in 2000 initiated adoption of ICTs as part of development planning, focusing on poverty reduction, improved governance, and better public services. MCTD, with support from INTIC, has the mandate to oversee and promote the revised Information Society Policy (2018).[54] Technical implementation of the Policy is the responsibility of INAGE, as the government digital transformation agency. The Policy addresses new technological challenges, aiming to enhance economic and social development and make Mozambique a nation where all citizens have access to and benefit from ICTs. It emphasizes the importance of e-governance to enhance public service delivery, increase transparency, and facilitate greater citizen participation in governmental processes.

The Information Society Policy is governed by six principles, and includes some key elements related to data governance. The six principles are *info-inclusion*, ensuring universal access to ICTs regardless of geographic location, age, gender, or social class; *integration*, aligning ICT strategies with national development goals; *articulation*, coordinated implementation across all relevant sectors and stakeholders; *sustainability*, ensuring long-term positive impacts of ICT investments; *information security*, guaranteeing availability, integrity, confidentiality, and authenticity of information; and *good governance*, promoting transparency, accountability, and efficient governance through ICTs. The Policy includes a goal

[54] República de Moçambique, Resolution 17/2018 from 21 June, 2018.

of establishing a robust regulatory framework to support ICT development and secure digital transactions while promoting a culture of information sharing and transparency, and mentions strengthening statistical capacity, data quality, and data availability. According to project documentation for the World Bank's Mozambique Digital Acceleration Project (MDAP), the strategy has "remained largely aspirational due to limited financing and implementation of key initiatives at scale"[55], and there do not seem to be other high-level public commitments to the data governance agenda.

Several relevant additional strategies currently under development include the following:

- A national digital transformation strategy for the economy is being prepared. MCTD, through INTIC, is leading its development, with financing from the World Bank's MDAP Project.
- The World Bank's EDGE project is financing the creation of a public sector digital transformation strategy led by INTIC.
- INTIC is collaborating with the World Bank (under the EDGE project) and UNESCO (which has developed a Recommendation on the Ethics of AI) to develop an AI strategy.
- INE is preparing an updated national statistical strategy with support from the Partnership in Statistics for Development in the 21st Century (PARIS21).

The creation of these strategies provides opportunities to mainstream attention to data governance and use across several high-level government priorities.

B. Policy/Legal Framework[56]

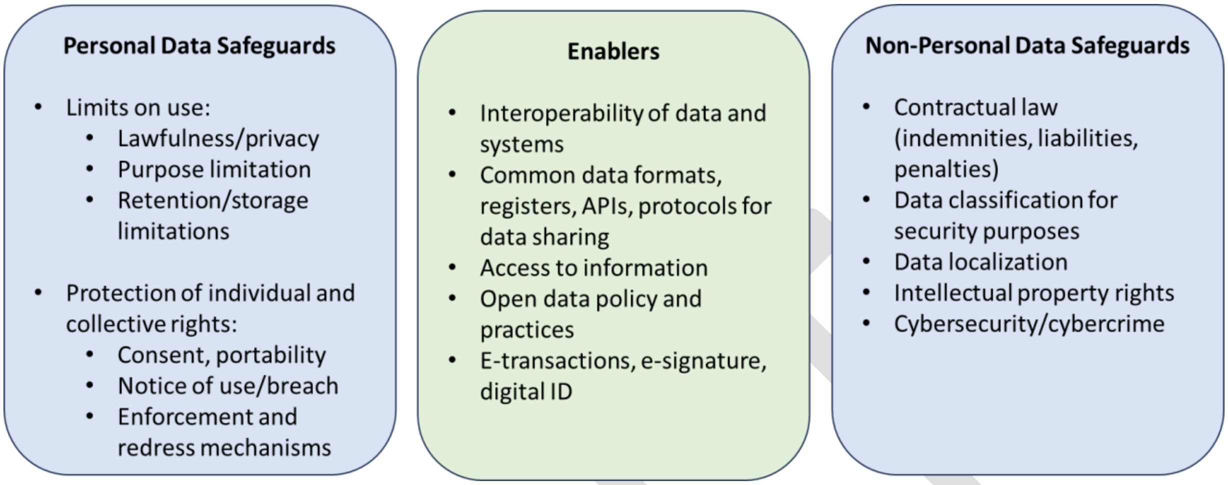
A comprehensive legal and regulatory framework for data includes enablers and safeguards related to data collection, management, use, and dissemination (Figure 3). This framework is only partially developed in Mozambique, with some ongoing developments in areas such as data protection/ownership, data regulations around e-commerce/transactions, right to information, and cybersecurity/cybercrime (beyond the scope of this report). Policy development in areas such as data sharing and interoperability is weaker.

I**n the absence of an overarching policy framework for data in Mozambique, different agencies have developed policies on an ad-hoc basis, leading to fragmentation and duplication of approaches**. For example, the Agência Nacional de Desenvolvimento Geo-espacial (ADE) has developed its own open data and sharing policy. A national policy for open data is now being considered.

[55] World Bank 2022a.

[56] The preliminary analysis and recommendations in this section are based on information and opinions collected from interviews undertaken and materials provided by the government and other local stakeholders during this study. This section is not based on detailed, legal due diligence and does not constitute legal advice. Accordingly, no inference should be drawn as to the completeness, adequacy, accuracy or suitability of the underlying Diagnostic, or recommendations, or any actions that might be undertaken resulting therefrom, regarding the enabling policy, legal or regulatory framework for Data in Mozambique. INTIC will benefit under MDAP from 1) the recruitment of a senior legal specialist, and 2) the recruitment of a law firm to advise on the legal and regulatory framework for digital transformation. This law firm will take on the necessary formal due diligence.

Figure 3. Overview of the elements of a comprehensive legal/regulatory framework for data governance



Source: *World Development Report 2021* team (adapted).

Currently, Mozambique lacks a general data protection law. Signaling its intent in this area, however, Mozambique is a signatory to the African Union's Malabo Convention on Cyber Security and Data Protection, which went into effect in 2023. In the interim, there are other legal sources that provide certain protections for the collection and processing of personal data, such as the Electronic Transactions Law of 2017.[57] The diagnostic team was unable to ascertain the extent to which implementation of these laws has progressed since the 2020 Digital Economy Diagnostic report, which found that "evidence suggests that the enforcement of these protections in the digital environment remains limited at best."[58]

In addition to its implications for domestic stakeholders, lack of comprehensive data protection undermines the potential for cross-border data flows, which in turn undermines Mozambique's efforts to strengthen its digital economy and increase regional economic integration. For example, Nigeria's High Court recently invalidated Nigeria's data sharing whitelist because it included Mozambique and other countries that it deemed to have insufficient data protections in place.[59]

The existing regulations do little to address data protection-related questions raised by the proliferation of new digital tools and platforms and the emergence of new technologies. According to the 2020 Mozambique *Digital Economy Diagnostic*, "New analytical and data collection methodologies raise several questions related to customer privacy rights and consumer protection. Data can be collected and shared with third parties without the knowledge of the data owner. There are currently no standard opt-in/opt-out policies for data sharing".[60] INE has created an internal strategy document to address concerns about big data and alternative data sources, aiming to create an enabling environment for data use while emphasizing the importance of capacity building and legal frameworks.

Due to the lack of national legislation, many government entities follow their own data protection policies and processes, which in some cases are explicit and in others may be ad hoc. The National

[57] DLA Piper 2024.
[58] World Bank 2020.
[59] Musoni, Karkare and Teevan, 2024.
[60] World Bank Group 2019.

Institute of Communications (INCM) has developed its own internal data use and protection regulations, which according to consultations for the diagnostic are based on the EU's General Data Protection Regulation (GDPR). The Bank of Mozambique has an internal policy on data security with processes to follow before sharing data and information with other stakeholders. Other entities that have internal procedures related to data protection include the National Institute of Health, the Human Health Research Authority, the National Medicines Regulatory Authority, and the Insurance Supervision Institute of Mozambique.

National efforts are now underway, led by MCTD, to strengthen data security measures through a proposed data protection law. The proposed Law on Personal Data Protection in Mozambique, now being drafted, will likely establish key principles such as consent, legality, purpose specification, accuracy, transparency, confidentiality, security, and proportionality.

There is also an intent to establish a National Data Protection Authority (ANPD) that would oversee compliance, issue guidelines, conduct audits, and impose sanctions. Organizations would be required to appoint Data Protection Officers (DPOs) to ensure adherence to legal requirements, manage data subject relations, and maintain confidentiality. The law is also expected to regulate international data transfers, requiring notification or authorization from the ANPD, depending on the receiving country's data protection standards. Non-compliance with the law could lead to severe penalties, including fines and imprisonment. The ANPD would be empowered to enforce these regulations and take legal action against violators. A legal review of the draft law will be conducted under the World Bank's MDAP Project.

Law No. 34/2014, enacted by the Assembly of the Republic of Mozambique, establishes the legal framework for the exercise of the right to information. This legislation is rooted in the constitutional principles ensuring democratic participation of citizens in public life and guaranteeing fundamental rights. The law details specific rights and duties regarding access to information:

- *Right to request information*—all citizens, legal entities, and media organizations can request public interest information from relevant entities.
- *Duty to disclose information*—public entities must proactively publish key information, including organizational structures, decisions affecting citizens, financial reports, audit results, environmental assessments, and public contracts.
- *Timely response*—requests for information must be addressed promptly, within a maximum of 21 days.
- *Free access*—information provision is generally free, except when reproduction, authentication, or certification incur costs.

Exceptions to the law are carved out for classified government information, confidential information (such as personal data), and information related to ongoing investigations or judicial matters. The law emphasizes the importance of open data and proactive disclosure of environmental and natural resource information especially to ensure transparency and good governance.

According to the Global Right to Information Rating, the law scores poorly relative to other countries due to its narrow scope (it only applies to the executive administration), broad exemptions, and absence of an oversight body that adjudicates refused requests.[61] According to the World Bank's *Systematic Country Diagnostic*, "Sufficient data on the performance of agencies legally bound to respond

[61] https://www.rti-rating.org/

to requests for information and effective feedback mechanisms will be key for ensuring the effective implementation of the right to information legislation, as well as to harness the potential of the media." The Mozambique *Digital Economy Diagnostic* also noted an implementation gap with respect to the Right to Information law.[62] Based on the consultations conducted for this diagnostic, it seems that it is not uncommon for public officials not to respond, or to take up to a year to respond to requests for information under the law.

In 2022, INE published the Microdata Dissemination and Access Policy, which aims to disseminate accurate and comprehensive statistical knowledge and promote its use among various stakeholders. It aims to guide institutions that are part of the National Statistical System in modern, timely, high-quality, and user-oriented dissemination practices, maximize the use of official data, and leverage new technologies for efficient data dissemination, all in the interest of supporting informed decision-making and contributing to national socio-economic development. It includes guidelines for ensuring accessibility, engaging with users to align data provision with their needs, presenting information clearly, guaranteeing simultaneous access to data for all users, and promptly rectifying any errors. Data privacy and security are emphasized through anonymization practices. Public use files provide anonymized data for general download, while licensed files offer less anonymized data under strict agreements, and local access files allow controlled access to sensitive data within INE premises. The diagnostic team was not able to ascertain to what extent the policy has been implemented.

Other policy areas that have a data governance component include Mozambique's Electronic Transactions Act (Law No 2/2017) which provides a general legal framework for the protection of ICT users in trade and investment, and covers electronic transactions, e-commerce, and e-government.[63] The Act legally established INTIC and charged INTIC with ensuring compliance with the Act. The Act includes provisions that cover electronic signatures, consumer protection in contracts related to e-commerce, the legal use of electronic services in public administration, and the Digital Certification System and Encryption, among other areas.

INTIC's recently approved Licensing Regulation[64] outlines requirements for the registration and licensing of electronic service intermediaries and digital platform operators in Mozambique, but some of its provisions are not fully aligned with international good practices. The aim of this regulatory framework is to foster a secure and efficient digital environment in Mozambique, promoting the growth of electronic services while emphasizing the importance of protecting user data and ensuring that platforms operate securely to safeguard user interests and national security. However, in practice there is a risk of potential negative impacts on the market, in particular because it imposes disproportionate financial burdens on digital platforms and burdensome requirements with a licensing regime when registration might have been more appropriate for building trust in the online environment. It will be critical for the implementation of any new requirements in this area to be aligned with the country's broader data governance objectives and framework, and reconsideration of some of the requirements is advisable (discussed further in the Recommendations section of the report).

[62] World Bank 2020.
[63] LexAfrica 2017.
[64] Regulation for the Registration and Licensing of Intermediary Providers of Electronic Services and Digital Platform Operators. Encapsulated in Decree No. 59/2023.

On the infrastructure side, the government is currently working on regulations regarding the construction and operation of data centers.

C. Institutional Structures, Responsibilities and Capabilities within Government

In the absence of a coordinating mechanism on data governance in Mozambique, institutions have pursued their own priorities in silos. This highly fragmented institutional environment does not support a coherent approach to data governance and management and leads to duplication of efforts and inconsistent application of data protocols. This section provides a synopsis of the key government institutions relevant to the data agenda.

In January 2025, a Presidential Decree mandated the reorganization of a number of government ministries.[65] For the data governance agenda, a significant change was the dissolution of the Ministry of Science, Technology, and Higher Education (MCTES) as well as the Ministry of Transport and Communications (MTC), and the creation of a new Ministry of Communications and Digital Transformation (MCTD). MCTD's portfolio includes the technology-related aspects that were under the purview of MCTES, plus the communications-related aspects that were under the MTC, and now brings several agencies and entities that had been under different ministries under the same institutional umbrella.

MCTD has the mandate to promote the advancement of information and communication technology (ICT) and is designated to provide leadership on the data and ICT agenda. The Minister of Communications and Digital Transformation oversees the following independent public agencies:

- The **National Institute of Information and Communication Technology (INTIC)** regulates, supervises, and monitors the ICT sector, and is responsible for ensuring compliance with certain regulations and promoting risk management practices and awareness programs. INTIC's roles and responsibilities include fostering competition, developing policies and standards for the security and integrity of computer systems, and proposing policies and strategies for the ICT sector, including on data protection and cybersecurity. INTIC has a mandate to design a data governance framework for the country, including an interoperability framework for the public sector (the scope of which is currently being revised). INTIC is also leading preparation of the national digital transformation strategy. However, based on the consultations undertaken for this diagnostic, INTIC has faced considerable challenges related to human and technical capacity, which have impeded its ability to execute its regulatory and oversight functions effectively. Previously, INTIC's position within the former MCTES limited its influence vis-à-vis other ministries, contributing to a leadership void in the broader data governance landscape. This lack of centralized leadership posed significant obstacles to advancing a cohesive and integrated approach to data governance within the public sector. Following the creation of MCTD, which is expected to have more leverage over the digital agenda than its predecessor, INTIC may be in a stronger position to enforce data standards across ministries, though it is still too soon to assess the impact of the reorganization.

- The **National Institute of Electronic Government, Public Institute (INAGE, IP)** coordinates and implements electronic government and ICT solutions with other public and private entities and civil society. INAGE has the mandate to implement e-government initiatives and is currently

[65] Decreto Presidencial n.º 1/2025, available at https://archive.gazettes.africa/archive/mz/2025/mz-government-gazette-series-i-supplement-dated-2025-01-16-no-12.pdf.

rolling out the new interoperability platform (mentioned above). INAGE has an engineering team that possesses skills in data science, software development, and cybersecurity to implement e-government services and manage data infrastructure but lacks a plan for AI solutions. INAGE also has a department of planning and cooperation. Interviewees noted they face staffing constraints and challenges, including a lack of advanced training opportunities and difficulties with staff retention given the availability of higher-paying opportunities elsewhere. They have been receiving support from ongoing World Bank projects and from the Eduardo Mondlane University (UEM) computer science department, and are seeking external partnerships from countries like South Africa and Portugal.

- Formerly under MTC and now part of MCTD, the **National Communications Institute of Mozambique (INCM)'**s mandate is to regulate the communications sector in the country. Its data-related functions are relatively well resourced with an in-house team comprising data analysts, data scientists and digital security experts who oversee their servers (also shared with other government entities). INCM generates dashboards using tools like PowerBI, Metabase, and Tableau; annually publishes a report to provide key updates and statistics; and has created an Interactive Voice Response (IVR) tool for people to submit notifications on disasters such as cyclones. It also collaborates with INE to conduct surveys regarding national communications and collaborates with ADE to create maps and visuals on network coverage. Retaining talent is a key challenge for INCM. Its strategies include supporting professional development by offering training opportunities and offering clear career progression paths.
- **Agência Nacional de Desenvolvimento Geo-espacial (ADE)**'s mandate is to promote the use of geospatial data among all institutions in Mozambique for data-driven decision making. ADE has technical staff with relevant data skills and a focus on retaining talent through training programs. The fact that ADE generates revenue from services helps bolster the resources it has available for staff salaries/recruitment. As an illustration of its technical capacity, ADE collaborates with various government agencies to develop tools for collecting geospatial data and platforms to host and manage geospatial data. ADE trained 50+ people from financial institutions and mobile service staff to collect geospatial data on a project sponsored by the Bank of Mozambique.

In 1996, a law was enacted creating the National Institute of Statistics (INE) and the National Statistical System (NSS), which is "responsible for the production and dissemination of official statistical information of general interest to the country".[66] The law is currently under revision (as of 2020). There is a High Council of Statistics (CSE) that oversees, guides and coordinates the NSS.[67] Within the NSS[68]:

- The **National Institute of Statistics (INE)** is the executive organ of the National Statistical System.[69] It plays a key role in implementing the government's data agenda by overseeing official statistical data production and dissemination and coordinating with ministries to delegate certain aspects of domain-specific data collection that will be inputs to official statistics (this does not include administrative data). In addition to its coordination and oversight of this decentralized system of data collection, INE staff provides minimal to high-touch technical assistance as per need and

[66] Instituto Nacional de Estatística 2024.
[67] United Nations Statistics Division 2024.
[68] There is also a Coordinating Council of the General Population and Housing Census (CCRGPH) according to UNSTATS. See United Nations Statistics Division 2024.
[69] United Nations Statistics Division 2024.

works closely on official statistical data production-related activities with counterparts at relevant agencies. The World Bank's National Statistics and Data for Development Project supported INE from 2018-2022, which allowed INE to substantially improve its technical capabilities and produce high-quality data, which have in turn been increasingly available for evidence-based policy formulation (see Box 2).[70] However, as INE's coordination only focuses on production and dissemination of official statistics, there is a need for a more holistic view of the supply of and demand for public data that includes administrative data, citizen-generated data, and other non-traditional data sources.

- The **Bank of Mozambique,** as an organ of the National Statistical System, has the mandate to centralize and compile the monetary and foreign exchange statistics that it deems necessary for the pursuit of sound policies. The Bank has a Department of Statistics and Reports that is mostly comprised of staff with training in economics, finance, and management, but also includes one statistician and one IT specialist. They have adopted the BSA (Bank Supervision Application) as a technical tool for data collection, which based on the consultations seems to meet their needs, and they use SAP for data storage. For data manipulation and analysis they use Excel, which they find less satisfactory. The Department is currently in the process of a database integration project that will streamline data collection (currently there is overlap/duplication) via a data warehouse, and enable use of tools like BI or SQL to manipulate data, rather than Excel. As part of this project, they contemplate creating a broader framework for data governance at the Bank of Mozambique. Many Central Banks in Africa are embarking on a similar database integration project and there may be opportunities for peer-to-peer experience sharing (e.g., Ivory Coast, Tunisia).

- **Eight ministries have been official delegates of INE** for data sharing for the SDGs. Prior to the ministry reorganization of January 2025, these included MCTES, Ministry of Education and Human Development, Ministry of Health (MISAU), Ministry of Agriculture and Food Security, Ministry of Labor, Employment and Social Security, Ministry of Public Works, Housing and Water Resources, Ministry of the Sea, Inland Waters and Fisheries, Ministry of Land and Environment. The former Ministry of Economy and Finance (MEF) chaired the SDG reference group.[71] Data sharing between INE and the ministries that are official delegates of INE is mandated by the statistical law.[72]

Box 2. INE's capacity has improved significantly in recent years

The World Bank's National Statistics and Data for Development Project (2018-2022; P162621) was instrumental in strengthening the technical and institutional capacity of INE and the NSS, facilitating training in capacity building, and equipping INE and its Provincial Delegations, including the National School of Statistics, with modern means of operation and communication (ICTs). The support allowed INE to produce a National Statistical Strategy, conduct a population and housing census, and implement several household surveys. The Implementation and Completion Report for the project credited government ownership and accountability around a strengthened NSS as key for the project's successes, noting that INE has demonstrated a high level of commitment to statistical development. The report did note that the project's Steering Committee could have done more to facilitate implementation and

[70] World Bank 2023a.
[71] https://unstats.un.org/capacity-development/UNSD-FCDO/mozambique/#:~:text=The%20National%20SDG%20Reference%20Group,group%20responsible%20for%20coordinating%20the
[72] https://unstats.un.org/capacity-development/UNSD-FCDO/mozambique/

advised that "the strength of simpler implementation arrangements with fewer implementing agencies should not be overlooked in order to maximize coordination and lead to the successful and efficient implementation of project activities."

Building on the strong results of the statistical project, Mozambique is now part of the World Bank's Southern African Development Community (SADC) Regional Statistics Capacity Building Project (P175731), which aims to 1) improve harmonization, quality, and dissemination of core social and economic statistics; and 2) close gaps in data production, statistical capacity, and data use.

Source: Drawn from World Bank 2023c.

Other key players relevant to the data ecosystem include the following:

- The **Ministry of Finance (MoF)** does not have an explicit mandate in terms of digital transformation and data governance in Mozambique. However, it plays a key role in the allocation of funding for agencies and IT systems, which determines their relative capacity to implement relevant digital initiatives. MoF also has strong convening power and is key in leading initiatives that require cross-government coordination. With respect to data, MoF has internal teams that analyze data for strategic decision-making, while relying on INE to process and release statistical information about the data they collect, e.g., family aggregates, financial data, etc.
- The **Center for the Development of Finance Information Systems (CEDSIF)** is responsible for developing and maintaining financial management information systems. CEDSIF is a public institute under the MoF, with financial and administrative autonomy.[73] It manages all Public Financial Management (PFM) systems and digital tasks relating to most public expenditure data in Mozambique. This includes collecting, storing, processing and auditing public financial data at the central and subnational government level. At the MoF's request, CEDSIF has also developed and hosts several non-PFM systems, and as a result holds significant administrative data.[74] CEDSIF reported 271 staff in total by end of 2023 and has the largest headcount of IT staff in government.
- The **Tax Authority (AT)** has three departments that work extensively with data: IT, Statistics, and Planning. In the course of its work, the AT analyzes data (for example, to determine whether audits are warranted), and shares (provides and receives) data with other government institutions including the customs authority, Ministry of the Economy, MoF, Bank of Mozambique, Revenue Authority, Ministry of Agriculture, Environment, and Fisheries, and others, as well as with commercial banks.

Table 3 provides an overview of the institutional landscape for data governance.

[73] Agencies with the status of "Public institutes" in Mozambique are able to generate revenues.
[74] These include for example the National State Human Resources Management System (e-SNGRHE) owned by the Ministry of State Administration (MAEFP), and the Management System for Beneficiaries of Basic Social Assistance Programs (SGB), owned by the National Institute of Social Action (INAS).

Table 3. Institutions and Actors that Perform Data Governance Functions in Mozambique (current landscape)

Thematic Clusters and Functions	Institutions and Actors
Strategic Planning • Developing strategies and policies in line with the social contract for data[75] • Establishing institutional arrangements • Coordinating between public, private and civil society entities	*Data governance arrangements* • MCTD (INTIC, INAGE, INCM) • INE
Rulemaking and Implementation • Legislating/regulating • Setting standards for data classification, management and sharing • Data management procedures (e.g., storage, security) • Providing clarification and guidance	*National legislature and subject-specific regulators* • INTIC • INCM • INAGE
Compliance • Enforcing (licensing, certification) • Auditing • Arbitrating • Remedying	*Watchdog and umpire* • INTIC (for electronic transactions) • INAGE (acts as a certifying body within the State's Electronic Certification System) • Data Protection Authority (envisioned) • INCM (compliance in the telecoms and postal sectors)
Learning and Evidence • Engaging in backward-looking monitoring and evaluation • Engaging in forward-looking learning and risk management	*Government representation* • INTIC • ADE *Knowledge Community* • Think tanks/NGOs • UEM • International development banks

Source: The structure for this table (including the left column) is taken from World Bank 2021a (lightly adapted), with the Mozambique-specific content added by the diagnostic team.

D. Government Data Management Policies, Procedures, and Data Availability

Data management practices vary across government agencies, with each ministry maintaining its own systems and procedures (which are not always documented) for data collection, management, exchange, and hosting. Reflecting the lack of a unified strategic vision and technical framework for data governance, multiple ministries (e.g., INCM[76], finance, health, justice, transport, internal affairs) have developed their own unique identification numbers/systems. There are concerns about data vulnerability

[75] Per World Bank 2021a, a "social contract for data" would "enable the use and reuse of data to create economic and social *value*, promote *equitable* opportunities to benefit from data, and foster citizens' *trust* that they will not be harmed by misuse of the data they provide" (World Bank 2021a).

[76] INCM plans to implement a biometric system for SIM card registration.

due to the lack of data security protocols and exacerbated by the absence of centralized data management or storage, and the lack of information sharing guidelines, combined with the absence of interoperability with the foundational ID system. Some entities rely on email or physical methods (i.e., flash drives) for data sharing and do not appear to have associated controls/procedures in place, though they may use encryption. Data sharing sometimes happens on demand (upon request from another Ministry) as opposed to via a set schedule or agreement.

Data production, data initiatives, and digital services

INE is responsible for the "rating, calculation, coordination and dissemination of the official statistical information of the country."[77] INE plays the central role in statistical coordination by delegating specific data collection responsibilities to other government entities ("delegated entities", i.e., sectoral ministries and agencies) and collaborating with these entities to ensure statistical data quality and compliance with principles of statistical confidentiality. Each year "the delegated entities of INE must submit… the respective plans for delegated statistical activities and the corresponding implementation reports," and they must submit the statistics they produce to INE for technical approval prior to data dissemination.[78] Agencies apply specific quality norms within their processes, and there are ongoing revisions to the National Quality Assurance Framework to ensure statistical data quality and adherence to international standards. However, the implementation of these frameworks and quality norms for data management are not consistently monitored or evaluated.

At the ministry or directorate level the government engages with international partners to align with international standards on data production. For example, the Directorate of Planning, Statistics & Cooperation (DPEC) of MCTD is part of annual meetings with the African Union on harmonization of statistics, and uses the outputs of those meetings to guide their methodology. However, data sharing between national and regional entities could be improved: "For instance, Mozambique's National Institute of Meteorology (INAM) produces weather forecasts to be distributed in the country, but because of delays in receiving parallel continental forecasts, there are two (sometimes unmatching) sets."[79]

The Directorate of Planning, Statistics & Cooperation (DPEC) within MCTD is an example of a government entity that collects data and works closely with INE. DPEC-MCTD has a Statistics Department that collects data in three domains relevant to the Ministry's work: higher education, R&D, and business innovation. INE generates the sample frames for the surveys that DPEC fields to various groups. As mentioned, DPEC also works to coordinate its methodology with international standards—DPEC receives assistance from the African Union for its work with surveys on research and development (R&D) and innovation and follows applicable international data classifications (e.g., UNESCO Manual for Higher Education) in the three different areas. Once collected, data are submitted by DPEC to INE, and also to the African Union. DPEC's data is stored locally on the department's computers for now, but INE has a new platform for data storage that DPEC may be able to use in the future. DPEC uses Kobo Toolbox for data collection, and SPSS and STATA for data analysis.

Aside from official statistics, there are varying levels of data-driven activities, reflecting stakeholders' interest, capacity and resource availability. Individual leaders at some ministries and departments champion the data agenda within their institutions and have created policies and procedures, and their own digital applications. For example, ADE's MozGIS private platform allows clients like the Bank of Mozambique, ministries such as MISAU, and others to use GIS data for spatial analysis and map

[77] https://www.ine.gov.mz/en/web/guest/sen
[78] Ibid.
[79] Musoni, Karkare and Teevan, 2024.

visualization and for developing other applications. INCM has developed a digital portal for consumers to report fraud and has computational models to detect fraud on an ongoing basis. Agencies such as FUNAE (Energy Fund), the Bank of Mozambique, the National Disasters Management Institute, Ministry of Health, Ministry of Finance, and the former MTC have shown significant activity in adopting data-driven tools for service delivery and in releasing their data. The Ministry of Health used data for making lockdown decisions during the COVID-19 pandemic, while the MTC used data to understand the flow of people between terminals. Similarly, the Ministry of Mineral Resources and Energy and Bank of Mozambique have utilized geospatial data for infrastructure expansion planning. However, some agencies are in the initial stages of implementing data-driven tools.

The government has made strides in modernizing data collection methods, including INE's adoption of digital platforms like tablets with GIS capabilities for surveys. The Bank of Mozambique uses BSA (Bank Supervision Application) for data collection, which allows different actors to upload data that can then be accessed by the Bank. Government efforts are underway to develop an integrated platform for profit declaration by companies, with collaboration among various agencies and ministries to streamline reporting processes and ensure data sharing for regulatory purposes, especially in sectors like tourism. Similarly, with funding from the MDAP project, INCM is designing a system to fully automate receipt of data and statistics from mobile network operators (MNOs), which will be stored on their own secure servers. INTIC is working with the Mozambique Association of App-Based Taxi Drivers (ATAM) to promote the registration and licensing of digital application entities for taxi services, as the use of these applications involves the collection of user data, such as location and destination, which can pose risks if data are not adequately protected.[80]

Despite challenges of data availability and governance, different parts of the government in Mozambique have developed and launched data-driven applications. A couple of examples include:

- Public facing systems such as the online passport application platform (SIGAV)[81] where citizens can apply for a new passport, and where foreigners can request visas and other travel and residency permits. Over 140,000 people[82] have requested services through the platform since its launch in 2020. Other examples include the biometric driving license and motor vehicle registration system.[83]
- Internal government systems such as the Electronic State Financial Administration System (e-SISTAFE), the Government Portal of Mozambique, and the Health Information System for Monitoring and Evaluation.

Interoperability

The lack of an operational interoperability framework creates duplication of efforts related to data collection and storage, software development, and licensing in Mozambique, making it difficult to effectively utilize shared assets.[84] In many cases systems have been installed via "development partner-supported initiatives without sufficient consideration to interoperability".[85] There is a Decree on electronic government interoperability to improve data exchange between systems, and INTIC is currently updating a proposed interoperability framework, which when approved will allow government entities to

[80] INTIC 2024.
[81] https://sigav.senami.gov.mz
[82] https://www.trade.gov/market-intelligence/mozambique-digital-governance
[83] 360 Mozambique 2022.
[84] World Bank 2021b.
[85] Musoni, Karkare and Teevan, 2024.

access data from other parts of the government more efficiently. INAGE is the implementer of the envisioned government interoperability platform, which leverages X-Road technology from Estonia, and is being designed and operationalized with support from the World Bank under the EDGE project. The verification of birth certificates and the extraction of personal details for ID issuance by the National Directorate of Civil Identification (DNIC) has been established as the primary use case for the interoperability platform at the outset. The platform will also support interoperability between IDs and the tax identifier (NUIT), inter alia.

An interoperability working group has been created, composed of INAGE, DNIC, DNRN and the Revenue Authority (AT).[86] A comprehensive work plan has been developed, as well as TORs for technical assistance. At the same time, INE has plans to transition to SDMX for data exchange[87] and intends to create a data warehouse to enhance data accessibility and utilization. The Bank of Mozambique also intends to adopt a data warehouse as part of its internal database integration project, and as part of this effort, contemplates creation of a high-level data governance framework for the Bank. There could be room for exchanging experiences and joining forces in this area.

Open data in Mozambique

The Government of Mozambique has taken steps to make official data available to the public and improved significantly on the Open Data Inventory (ODIN)[88] from 2020 (ranked 174th out of 187 countries) to 2022 (ranked 151st out of 195 countries). Still, there is substantial room to continue building on these efforts. For reference, stronger performers in the region include Uganda and Kenya, which rank 103rd and 104th, respectively. Looking at the ODIN more closely, Mozambique has an openness score of 29 out of 100 and coverage score of 46 out of 100. Mozambique's data coverage is very good in some areas including Money and banking; Balance of Payments; and Crime and justice; whereas data coverage is poor in areas like Poverty and income; Energy; and Pollution. The openness score is relatively consistent across the sectors. Table 4 presents data on the availability of SDG indicators across some comparator countries, which show that Mozambique is similar in this regard to others in Sub-Saharan Africa, and the countries' overall ODIN scores.

Table 4. Indicators of Data Availability in Mozambique and Select Comparator Countries

	SDG Data Availability*	ODIN Overall Score 2022**
Mozambique	**44.9%**	**36.8**
Botswana	49.2%	51.2
Ethiopia	44.7%	40.7
Kenya	50.9%	45.5
Madagascar	48.4%	33.1
Malawi	48.8%	38.3
Nigeria	44.6%	44.5
Rwanda	44.6%	59.4
South Africa	53.2%	52.6
Uganda	52.5%	45.6

[86] Bhatti 2024.
[87] With support from the African Development Bank.
[88] https://odin.opendatawatch.com/Report/countryProfileUpdated/MOZ?year=2020

Zimbabwe	42.9%	42.3

*Proportion of series with data for at least 2 years since 2015. Source: Goessmann, Idele, and Jauer, et al., 2023.
**Open Data Inventory (ODIN) 2023.

The government shares data through various outlets, including its open data portal[89], but ensuring consistent and comprehensive updates would improve the usefulness of public data. The Bank of Mozambique publishes macroeconomic data regularly on its website. Mozambique is a member of the Extractive Industries Transparency Initiative (EITI) and has achieved a moderate overall score (82.5 points) in EITI implementation, according to the Board in 2023, "having contributed to enhanced transparency in both the mining and the oil and gas sectors". [90] The country's score on the Open Budget Index, however, was 47 out of 100 in 2023, which according to the International Budget Partnership indicates that it is not publishing enough information to support informed public debate.[91]

E. Demand for Data and Data Services, and Civic Engagement

There is a growing interest in and utilization of government data by various stakeholders in Mozambique, reflected in demand for data and data services, as well as civic engagement more generally. Local businesses, academics, and universities use government data for diverse purposes. For example, companies in sectors like finance utilize government data for fraud detection through machine learning models; students and academicians use government data for research; the media relies on government data for reporting; etc. However, there are challenges in accessing and utilizing government data due to the lack of digital infrastructure and data availability in digital formats.

While the government has established some programs for civic engagement and outreach related to data and data-driven products, stakeholders mentioned that the outreach efforts are insufficient. INE for instance engages with media professionals on WhatsApp to provide them with data in desired formats. The Ministry of Health relies on press conferences and press statements to inform the public, without regularly publishing data in parallel for the public to access and analyze. There is no formal mechanism to assess the demand for data from a broad spectrum of stakeholders and data users, and as a consequence, there is no reliable indication on priorities for the data ecosystem—the system relies only on partial and unverified information.

There are also broader concerns about civic space. The World Bank Systematic Country Diagnostic noted that "some earlier gains in freedom of expression and assembly seem to have been recently reversed, leading CIVICUS Monitor to change its rating for its civic space from "obstructed" in 2020 to "repressed" since 2021.[92] Freedom House rates Mozambique "Partly Free", with a score of 12/40 on Political Rights and 29/60 on Civil Liberties.[93]

There is demand for the government to make updated and accurate statistics across various sectors available. DPEC (within MCTD) mentioned that they sometimes get calls from students or researchers requesting data, which DPEC will provide, but they do not track statistics on visitors to their website,

[89] https://mozambique.opendataforafrica.org
[90] EITI 2023.
[91] https://internationalbudget.org/open-budget-survey/country-results/2023/mozambique#:~:text=How%20comprehensive%20is%20the%20content,makes%20available%20to%20the%20public%3F&text=Mozambique's%20transparency%20score%20of%2047,near%20its%20score%20in%202021.
[92] https://monitor.civicus.org/
[93] https://freedomhouse.org/countries/freedom-world/scores

downloads, or other indicators that would create a picture of the public demand for their outputs. They mentioned data dissemination and the lack of feedback loops with data users as challenges for the department.

Obtaining data from the government that is not already posted online can be a challenge. Based on the consultations, it seems that the process for students, for example, to request data or reports from public agencies typically involves formal requests through universities, with varying timelines (3 months to a year) for data retrieval. Data is usually shared in .CSV or Excel formats. INCM receives data requests from various national and international/development partners, such as data on the number of licenses or Internet Service Providers (ISPs). They are willing to share data analysis and results, but they host data in national servers and do not want it to leave the country.

The private sector could benefit from increased access to public data. The project documentation for the EDGE project notes that "the absence of open standards, open data and open software policies tend to crowd-out, rather than support, private sector opportunities" in Mozambique. The government operates an online procurement portal[94], but an assessment conducted by an outside group suggests that there is significant room to improve the transparency of the procurement process.[95]

F. Funding a Data Program

The Government of Mozambique has identified digitalization as a strategic priority but has yet to adequately mobilize resources to operationalize its data and digital transformation agenda. Some key takeaways from the consultations include:

- **National funding for data programs in Mozambique is limited and primarily comes from external donor support.**
- **The National Directorate of Geo-Spatial Information (ADE), designed as a state-owned enterprise, initially received funding from the World Bank and from the government, and now mainly relies on generating its own revenue (ADE is still receiving some limited funding via the MDAP project).** ADE's mandate includes promoting the use of geospatial data and managing a GIS platform for storing data from different departments. Private access to the platform involves annual fees, providing access to data, tools for map creation, and applications. ADE now operates on a mixed funding model and is two-thirds self-financed and one-third externally financed.
- **Ministries often use internal resources or funds from donors to develop applications or e-services.** Funding for ICT infrastructure and staff skill development is essential for managing a data program effectively. The government intends to move toward common infrastructure solutions across agencies, to be managed via INAGE (including management of the government communication network, government data center, and an IT equipment inventory for all ministries), but INAGE does not currently have the capacity to fulfill this mandate.
- **Public-private partnerships also play a crucial role in data-driven solutions.** ADE, for example, collaborates with Eduardo Mondlane University to access research and data collection capacity. Similarly, the government has conducted hackathons to develop solutions for the public sector. However, more partnerships are needed to leverage external expertise and resources for data-driven initiatives.

[94] https://www.ufsa.gov.mz/concursos.php
[95] https://www.tpp-rating.org/page/eng/country/Mozambique/2020

G. Technology and Skills Infrastructure

In terms of physical infrastructure, connectivity and digital infrastructure are relatively limited in the country. Internet penetration was 21 percent in 2023, and 45 percent of individuals owned a mobile phone.[96] There are 27 unique mobile-broadband subscriptions per 100 inhabitants.[97] With a literacy rate of 60 percent[98], a substantial portion of the population has minimal familiarity with the Internet. Per the ITU's state of "Universal and Meaningful Connectivity" in Mozambique, connectivity is limited overall, but Mozambique has strengths in gender parity in connectivity and mobile broadband cost.[99] Recent data suggests that mobile connections and social media use have been increasingly rapidly since 2023.[100] Companies like Amazon Web Services (AWS), Heroku, and Microsoft Azure provide cloud services in the country[101], and there are new private data centers in Mozambique (iColo and Raxio).

A key effort to foster Mozambique's digital infrastructure was led by the former Minister of MCTES, who helped create the National Company of Science and Technology Parks (ENPCT) in 2012 to foster training and innovation.[102] This founding vision has not been fully realized. The initiative aimed to create an ecosystem conducive to data and ICT-led business development and included establishment of a national data center in Maluana. According to a 2024 European Union-funded report on cross-border data flows in Africa, in Mozambique the government's interest in data centers grew out of a view that data sovereignty means data localization, and hence there was a drive to ensure data could be stored in the country.[103] Now the data center does not receive sufficient funding on a reliable basis.[104] Operational and financing challenges mean that many government agencies opt to host their data elsewhere[105], but the government is actively trying to increase the data center's utilization rate, which is currently around 50 percent.[106]

The government has created a Computer Emergency Response Team (CSIRT) with a Security Operations Center (SOC) to protect the public sector (in particular, critical infrastructure assets) and to handle aspects related to information security and promote the security culture. However, its capacity is weak—strengthening the CSIRT is a key focus of MDAP's support on cybersecurity.

Despite the nascent stage of digitalization in Mozambique, the demand for high-quality digital skills in the country is expected to grow significantly. A World Bank Group study[107] has found that by 2030, digital skills will be required for 20-25 percent of jobs in Mozambique, or the equivalent of about 3.7 million workers, across various sectors of the economy. Most of these jobs are likely to require only foundational digital skills but the skill gap is real. It is more pronounced for high-end jobs, which while likely to be lower

[96] https://www.itu.int/hub/publication/D-IND-ICT_MDD-2023-2/
[97] Source: "Unique" subscribers calculated as GSMA Mobile Broadband Capable Connections / GSMA SIMs Per Unique Subscriber / United Nations Population.
[98] https://databank.worldbank.org/source/world-development-indicators
[99] https://datahub.itu.int/dashboards/umc/?e=MOZ
[100] https://datareportal.com/reports/digital-2024-mozambique
[101] https://www.trade.gov/market-intelligence/mozambiques-digital-transformation
[102] As of the January 2025 ministry reorganization, ENPCT is part of MCTD.
[103] Musoni, Karkare and Teevan, 2024.
[104] Despite these constraints, several projects have been undertaken in areas such as solar computing, 3D printing, modular housing, and waste management.
[105] For example, per the consultations for this report, MISAU is not confident relying on the national data center for this reason and stores its data elsewhere. Because of this, the Ministry effectively loses ownership of the data as there is no regulation stating that the data belongs to the government even when stored elsewhere.
[106] Musoni, Karkare and Teevan, 2024.
[107] IFC 2021.

in number can help make significant productivity gains in sectors such as the oil and gas, ICT, and e-commerce sectors.

While high quality data about digital literacy in Mozambique is scarce (which itself may be considered a proxy indicator), examples suggest low levels of data and digital capacity in Mozambique. Less than a third of the men in Beira (15 percent according to one source[108]) have even basic digital skills (the women fared only half as well). The levels are only slightly higher in Maputo, despite somewhat higher internet access in that city.

Mozambique's formal education sector is ill-prepared to meet the demand for digital capacity development. There is limited data about the quality and quantity of students in the formal education system with strong and marketable technical degrees. According to one source[109], none of Mozambique's universities offering computer science degrees rank among the top 150 in Africa or the top 1000 in the world. This is unsurprising given that Mozambique's educational levels are among the lowest in the world (fewer than 4 percent of the population receives secondary education, and at 26 percent Mozambique's gross enrollment in secondary education is the lowest in Southern Africa). This deficit hinders industry growth and innovation, and it similarly impedes the progress of regulatory bodies. The talent gap slows the ability of regulatory agencies to build their internal capacities and gain in-depth knowledge of the digital ecosystem, which is crucial for creating forward-looking regulations for emerging technologies, including AI. Such regulations are essential to attract investment and foster innovation. For regulators, it is vital to have strong internal capabilities to understand the commercial needs of ICT players and how these needs interact with regulatory frameworks.

Digital capability within the government is uneven, with a few agencies leading the way while most others lag behind. Unsurprisingly, entities such as the National Institute of Electronic Government, Public Institute (INAGE, IP) which has the mandate to implement the government's digital transformation strategy (once developed), and Agência Nacional de Desenvolvimento Geo-espacial (ADE) which has the mandate to promote the use of geospatial data among all institutions in Mozambique, include engineering teams that possess skills in data science, software development, and cybersecurity, but they too struggle to hire and retain technically skilled staff. Many other government agencies still use basic tools such as Excel for data analysis and even then face capacity and skill challenges.

Ministries and agencies typically have in-house capacity for IT and data management functions, though it may be limited. Overall, data skills and capabilities within the government are unevenly distributed. While many ministries tend to rely on tools like Excel for data analysis, highlighting a need for investment in advanced data analytics tools and training programs, a few agencies use tools like Python, PowerBI, Metabase, and Tableau. Many of the agencies consulted mentioned that they face considerable financing and staffing constraints in the data domain, and some expressed a desire for training and capacity building opportunities. In some cases when internal capacity is insufficient, government outsources certain projects to private sector firms such as local firms (e.g., Technoserve) and international (e.g., the hiring of a Portuguese company by INE for developing a data collection app).

Overall, there is a need for more comprehensive training initiatives and capacity-building programs to enhance data literacy and analytical skills within the government workforce. While there are no dedicated national data training centers or institutes for civil servants, efforts have been made in the past to establish such institutions, like the Statistical School, which closed in 2022 due to low demand. The World Bank's SADC Regional Statistics Capacity Building project supports INE and its provincial offices

[108] https://www.equalsintech.org/casestudies-dsh-1/muva-tech:-bridging-the-digital-gender-gap-in-mozambique
[109] https://edurank.org/cs/mz/

through the procurement and upgrading of ICT equipment and software and, inter alia: 1) training of technical staff; 2) strengthening INE's ability to deliver statistical training programs on: (i) data processing and analysis; (ii) modern tools for documenting and electronic archiving of data; and (iii) creation and management of databases and libraries; 3) supporting recruitment of senior consultants to be stationed in provincial offices; 4) improvement of the national data archive; and 5) recruiting graphic designers.[110]

Personnel-related issues also constrain the government's ability to recruit and hire staff to fill gaps. One factor is the implementation of restrictions on hiring new civil servants that have been in place for more than five years and have made it difficult for the ICT sector (ministries and regulatory authorities) to add new staff. Additionally, uncompetitive remuneration packages for public servants with ICT backgrounds undermine recruitment, since public servants get paid significantly less than their counterparts in the private sector, creating a talent gap.

Civil society and the private sector display similarly uneven levels of digital capacity within the country, with a few pockets of excellence unable to mask the largely low levels of data and digital capacity. At one end are companies such as Jon AI that claim to be developing AI solutions for the African market, but the bulk of the private sector is still "analog" with most companies not employing any IT personnel at all or being digitally active in any meaningful way. This deficit hinders industry growth and innovation. It is also unclear that there are many civil society organizations focused on data or digital issues or with the ability to use data effectively in pursuit of their objectives.

The World Bank, other donor organizations, and the private sector have developed various initiatives to address data capacity challenges in Mozambique. These include, in the public sector, initiatives such as the Centros Multimédia Comunitários (CMCs) and Centros Provinciais de Recursos Digitais (CPRD), which offer connectivity and ICT training in the provinces.[111] Notable initiatives within the private sector to address digital capacity gaps include programs such as MUVA's digital skills training[112] aimed at women in select urban areas, and the USAID-funded Alcancar digital training program[113] that targets frontline health care workers. Moreover, collaboration and partnerships between the government, academia, and the private sector are gradually emerging. For instance, there are partnerships between INCM and international telecom regulators and companies to develop training programs.

Despite prevailing challenges particularly regarding infrastructure and energy access outside urban areas, there is a growing number of students pursuing training in ICT-related fields in all provinces in the country. The decentralization of job opportunities has enabled career development even in provincial areas, fostering entrepreneurship and innovation. The talent pipeline for data skills through universities or vocational trainings is very limited.

Overall, while Mozambique faces infrastructure and capacity challenges, there is a growing IT industry and a demand for skilled professionals in data management and analysis. The ICT association, AMPETIC, represents a sizable portion of ICT professionals and companies. These companies in the private sector play a role in training new graduates and building civic capacity to use data safely and responsibly. Lastly, while universities like Eduardo Mondlane University (UEM) offer courses in statistics, economics, management informatics, computer science, and engineering, the consultations highlighted the need for a dedicated data science and data analytics program in the country.

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[110] LEGKL 2024.

[111] MDAP has $20 million allocated to digital literacy and skills programs to reinforce some of these initiatives.

[112] https://www.equalsintech.org/casestudies-dsh-1/muva-tech:-bridging-the-digital-gender-gap-in-mozambique

[113] https://viamo.io/global-health/alcancar-mozambique-maternal-child-healthcare/

H. Artificial Intelligence and Machine Learning

The Government of Mozambique's utilization of artificial intelligence (AI) technologies is limited, with ongoing work primarily in collaboration with universities and industries. The government's experience with big data deployment includes some initiatives such as INCM's use of AI and machine learning to detect SMS fraud, with over 80 cases detected daily. INCM is thus using machine learning models to bolster security by detecting and responding to phishing threats in real time, helping to protect critical infrastructure and information systems.

There is a growing interest in utilizing machine learning techniques, particularly in fields such as telecommunications. Looking ahead, there could also be opportunities to leverage AI in defense, robotics, education, and consultancy (one person consulted mentioned ideas such as creation of hybrid automation systems, virtual laboratories, chatbots, etc.). Mozambique attended the first UNESCO-Southern Africa sub-Regional Forum on Artificial Intelligence (SARFAI), an event hosted by Namibia that may reflect some momentum toward increased regional collaboration.[114] In terms of other disruptive technologies, the use of cloud infrastructure and "Internet of Things" (IoT) devices is likewise still nascent.

A report on AI Readiness from Oxford Insights examines the readiness of governments "to implement AI in the delivery of public services to their citizens".[115] The assessment looks at three pillars: 1) government (strategic vision, regulations, and internal digital capacity); 2) technology (maturity of the country's technology sector); and 3) data and infrastructure (data availability, representativeness, and infrastructure). Per the assessment, Mozambique's weakest of the three is its technology sector, with a score of 18 out of 100, followed by the government pillar, with a score of 22 out of 100. Data and infrastructure are much stronger, with a score of 37 out of 100.

Recommendations in this report do not address AI specifically. The report focuses rather on foundations that are essential to safe and effective use of AI, including data protection and digital transformation, as well as data production and availability. As noted earlier, the government is currently preparing an AI strategy. The National Artificial Intelligence Strategy will be developed and implemented under the umbrella of the Information Society Policy that ends in 2028.

3. Strategic action steps

Grounded in the ODRA Methodology, findings of the in-country consultations, and desk research on the current state of data in Mozambique, the report presents five sections of recommendations to strengthen data governance in Mozambique. Each section includes a brief overview of the issue area building on the main findings discussed in the previous section, recommendations for consideration, and relevant lessons from international experience.

The recommendations aim to build upon existing institutional arrangements and programs in data and digital transformation. Importantly, the recommendations are intended to prevent duplication of effort, avoid friction with the enabling political environment, and avoid creating structures that would increase administrative burdens. The recommendations, while broad, also aim to be selective, identifying priorities for the short- to medium-term. This recognizes that building a strong data governance framework is a process that happens over time and in phases, following a maturity model approach. The government and

[114] Oxford Insights 2023.
[115] Oxford Insights 2023.

its partners in Mozambique, with their in-depth understanding of the national context and challenges, should utilize these recommendations in the manner they deem most appropriate.

In January 2025, the Government launched the National Observatory of the Information Society (O-SIM), a new mechanism to measure progress on the "development of the information society" in Mozambique.[116] By defining a consolidated set of official ICT indicators for the purposes of decision making and international reporting obligations, the O-SIM will help the government track progress on a number of important areas relevant to the data governance agenda.

1. National data strategy and senior leadership

Currently, the government is preparing several national strategies with relevance to the data agenda, which creates an opportunity for the elaboration of a strategic approach to data and data governance. As a key enabler for digital transformation, a vision for a comprehensive data governance framework could be subsumed under the umbrella of the national strategy for digital transformation. Other strategies could then refer to and provide further detail on this overarching framework as needed for their more specific purposes. Establishing a common vision for the data agenda in the strategies under development would contribute to consolidating institutional goals, policies, and approaches across the government.

Specific recommendations for consideration are as follows:

a. **Establish an Inter-Ministerial Digital Transformation Steering Committee to strengthen coordination.** An Inter-Ministerial Digital Transformation Steering Committee could provide leadership and strategic direction on the digital transformation agenda broadly, including data governance and management. The World Bank or AfDB could play a convening role to assist with the formation of the committee. At the same time, it will be important to streamline governance structures and adopt integrated approaches to oversight and decision-making and avoid a proliferation of new government committees related to the digital agenda. Examples such as Estonia and Rwanda demonstrate the successful merging of overlapping digital governance bodies into a single, high-level committee or council (with sub-groups as needed) to oversee all digital initiatives under one umbrella. The committee could potentially serve as a transitional arrangement with its role being institutionalized appropriately after some time.

b. **Through a multistakeholder, consultative process led by the Steering Committee, establish a shared vision for the role of data in the digital ecosystem and articulate its fundamental role in achieving national objectives.** It is recommended that this vision be embedded within the government's new ***national digital transformation strategy***. The digital transformation strategy is currently under development and could articulate a national vision for the data agenda as an enabler of the transition to a digital economy as well as supporting more informed decision-making and policy/program implementation and monitoring by the government. The strategy should take a user-centric, problem-driven (rather than supply-driven) approach that addresses the questions: who is using data now, and how? What challenges do they face in leveraging data effectively and ensuring data protection and security, and how can government help address those challenges? The strategy would also outline the data governance framework including laws

[116] Details about the National Observatory of the Information Society (O-SIM), as well as the book on Measuring the Information Society (Systematization of ICT Indicators on Mozambique 2017 – 2024), are available at http://osim.gov.mz.

and policies already in place as well as articulating the elements that will be put in place over the medium to long term. The national digital transformation strategy would then become the high-level reference point on data and data governance across government and throughout the economic system.

c. **Ensure all strategies that play a role in advancing the data agenda are aligned with and refer to the overarching framework enshrined in the national digital transformation strategy.** Stakeholders crafting the ***new national statistical strategy*** have indicated that they envision developing standards for electronic data exchange and sharing. It is recommended that this be coordinated with the effort to create the new ***public sector digital transformation strategy***. While the national digital transformation strategy outlines the vision and parameters for data use and management across the economy, the public sector digital transformation strategy would articulate how the public sector will facilitate data sharing and reuse across government entities in the context of that framework. This would include official statistics developed by the National Statistical System led by INE, as well as all other data produced or leveraged by the government, including administrative data and data from non-traditional sources (big data, citizen-generated data, etc.). Common data standards across government and a holistic view of available data will enable the country to harness data's potential and ensure that data management practices are aligned with national development priorities. This is also an important enabler of the transition to digital, data-driven government services. More generally, a collaborative, engagement-driven model for developing and delivering data-driven services that could be outlined in the public sector digital transformation strategy would include strategies such as data sharing, co-development and collaboration, representation and voice channels, and continuity (versus ad hoc or sporadic initiatives). Finally, the new ***Artificial Intelligence (AI) strategy*** should emphasize the fundamental importance of 1) the availability of quality, comparable data as an input to AI; and 2) a robust data governance framework as an underpinning for safe and ethical use of AI. As such, the AI strategy should be aligned with these other strategies.

d. **Establish mechanisms for regular engagement with data users, including government analysts, the private sector, academia, the media, and civil society, to surface challenges and priorities within the data agenda.** By bringing together a range of stakeholders, the state of the data ecosystem will come into sharper focus and the government will have a better understanding of the needs of current and potential data users. A consultative approach can include partnerships with data intermediaries with deep connections within the communities targeted by specific government services.[117] This engagement will help the government foster a collaborative data ecosystem.

e. **As part of a demand-driven strategy for investing in data, identify high-value data use cases to demonstrate the value proposition for data in Mozambique.** Mozambique is at an early stage of digital development where it must prioritize data-related investments that have the potential to scale beyond their immediate areas of impact and that can serve as either foundations or enablers for other programs or services. One useful starting point is the recent EU list of what it calls "high-value datasets" held by the public sector[118] in data categories such as geospatial, earth observation and environment, meteorological, statistics, companies, and mobility. The list also includes core data attributes and publishing format suggestions. It would be ideal to identify several high-value data use cases to help take strategic planning from the abstract (data is valuable) to the concrete (data can help the country solve a defined problem), in collaboration with non-government actors including the private sector and civil society. Identifying use cases that line ministries see as valuable to their work can also be a way to create positive incentives for cooperation among government agencies, rather than relying solely on mandates and directives.

Lessons from International Experience

Governments are increasingly thinking about data as a strategic asset that can help them achieve their development goals—and as a critical asset, it needs to be managed in a coordinated, strategic way. Overarching national data strategies (as opposed to the institutional strategies of National Statistical Offices) are increasingly common. There are many examples of countries that have produced or are in the process of creating a national data strategy. For example, Uganda's National Data Strategy calls for data to be used in an innovative and responsible way to transform Uganda's development, and to maximize the value of data through governance, training, awareness and trust building.[119] Other examples from Africa include Ghana[120], Nigeria[121], Senegal[122], and Zimbabwe[123]. In addition to overarching data strategies, countries have more narrowly defined strategies as well, such as Uganda's Big Data Utilization Strategy[124], and specific strategies for AI and other disruptive technologies are proliferating (see Box 3). Thirteen countries (including seven African countries) are parties to the Global Partnership for Sustainable Development Data's Inclusive Data Charter[125], which aims to advance the availability and use of inclusive and disaggregated data.

At the same time, it is prudent to avoid proliferation of strategies and rather to focus on the substance of what the government wants to achieve. In some cases, countries have produced data strategies through a process largely driven by external partners, without necessarily securing buy-in from all national stakeholders. As a result, these strategies exist but have not become the kind of national reference that was intended. The recommendation to embed Mozambique's data strategy within the digital transformation strategy is intended to help avoid this outcome.

Box 3. Emerging Strategic Areas: Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) holds immense potential to propel low- and middle- income countries forward in crucial areas like agriculture, health care, education, energy, financial inclusion, climate resilience, and insurability in both the public and private sectors.[126] The African Union has recognized AI as pivotal to its Digital Transformation Strategy for Africa (2020-2030) and emphasized the need for coordination to implement a harmonized and interoperable legal framework to harness AI's potential across the continent. In July 2024, the African Union Executive Council endorsed the Continental AI Strategy, which

[119] https://www.unglobalpulse.org/project/national-data-strategy-uganda-looks-to-data-to-transform-its-economy/
[120] https://datapopalliance.org/projects/development-of-ghanas-national-data-strategy/
[121] NITDA 2022.
[122] Republique du Sénégal 2018.
[123] Research ICT Africa 2023.
[124] Ministry of ICT Uganda 2023.
[125] https://www.data4sdgs.org/initiatives/inclusive-data-charter
[126] World Bank 2024.

"underscores Africa's commitment to an Africa-centric, development-focused approach to AI, promoting ethical, responsible, and equitable practices".[127]

At the country level, Mauritius, South Africa and Rwanda score highest on AI readiness in Sub-Saharan Africa according to an index compiled by Oxford Insights.[128] Mauritius adopted its AI strategy in 2018[129] and has gone as far as enacting AI-specific legislation, having implemented licensing procedures for entities that provide investment and portfolio management services enabled by AI in 2021.[130] South Africa has developed a National AI Policy Framework focusing on leveraging AI for economic growth and societal well-being, paving the way for creation of a full AI Policy and aiming to position South Africa as a leader in the AI space.[131] Rwanda has set up a center of excellence for digitalization and AI called the Centre for the Fourth Industrial Revolution (C4IR Rwanda), highlighting its commitment to leveraging AI for economic growth.

While AI strategies and policies are important, the development of robust data protection and privacy laws, cybersecurity laws, intellectual property laws, and governance frameworks for cross-border data flows are foundational to ensure responsible and inclusive AI deployment. Additionally, it is essential to focus on skill development initiatives and workforce preparation for the transitions that AI will induce, and on the availability of locally relevant data to feed into AI systems. By strengthening foundational enablers and accelerating AI adoption and regulation, Africa can fully harness AI's benefits to drive sustainable development and improve quality of life for its people.

Around the world, governments often have a supply-side mindset when it comes to data initiatives. Government agencies frequently fail to monitor the use of data resources and are surprised when they learn that the data they have made publicly available is of little interest or use to citizens or businesses. Citizen engagement initiatives can fail, even when citizen participation is built into a program. For example, one study found little evidence of affected populations playing a significant role in the design or management of predictive analytics in the humanitarian sector.[132]

Identifying specific use cases for which there is data demand can be an effective way to start incorporating data use into government operations more strategically and effectively (see Box 4). There are many examples of successful government data initiatives that can be instructive for Mozambique. For instance, a forthcoming OECD study estimates that "the innovative use of open municipal data on transport patterns in London… enabled the development of innovative new start-ups and applications, including those that mixed publicly collected data with other sources to allow multi-modal analysis. Firm-level studies estimate that open data added GBP 12 million to GBP 15 million per year for firms.''[133] Sometimes, even relatively straightforward and basic data initiatives can yield powerful results. A recent study from Pakistan found that simply replacing paper records with an app-based electronic equivalent reduced doctor absenteeism substantially.[134] In Nigeria, Lagos state has automated several public services

[127] https://au.int/en/documents/20240809/continental-artificial-intelligence-strategy
[128] https://oxfordinsights.com/ai-readiness/ai-readiness-index/
[129] https://www.unesco.org/en/articles/mauritius-hosts-inaugural-ai-summit
[130] ALT Advisory 2022.
[131] Available at https://fwblaw.co.za/wp-content/uploads/2024/10/South-Africa-National-AI-Policy-Framework-1.pdf
[132] Hernandez and Roberts 2020.
[133] OECD, forthcoming.
[134] Rezaee 2022.

including land management, payroll management, and internal revenue management and demonstrably improved their efficiency and effectiveness.[135]

Box 4. Data Use Cases as a Means to Pilot New Approaches to Data Governance

Identifying one or two data use cases to pilot new procedures and approaches could help the Government of Mozambique refine its data governance framework. Data-driven interventions may include the application of "big data" (vast datasets derived from multiple public and private sources, collected in close to real time); AI/machine learning/algorithmic decision-making techniques; and statistical modeling (e.g., to develop scenarios that anticipate events).

A multi-stakeholder, consultative approach is essential to 1) identify use cases for which there is clear demand and immediate application(s); 2) design an approach that is feasible given the enabling environment; and 3) ensure that the case yields useful lessons for the elaboration of the data governance framework.

While each data initiative or tool will have unique features, some common aspects can help the government to identify and design high-value use cases:

- **Data flows across organizational boundaries within the government, so that different stakeholders can add to or access the same data.** For example, in a remote sensing example, one government agency may handle data collection by installing and maintaining cameras, another agency may provide the relevant geographical classifications to organize the data, and another agency may analyze and use the resulting data.
- **Tools are created with the interests of the ultimate user, the citizen, in mind.** There is a growing emphasis on creating tools that average citizens can use to make sense of data and make appropriate decisions. A hazard risk database in the US, for instance, includes powerful visualization and filtering tools for citizens to gather and track highly granular local data across a wide variety of parameters.
- **Experimentation and evolution.** Data-driven tools, despite their seeming ubiquity, are still new and their creators are still experimenting with their design and validating their effectiveness. Widely deployed sentiment analysis tools are a useful example of an approach that has thrown up unanticipated challenges related to disinformation, prioritization, and iniquity.
- **A premium on the user experience and safeguards.** Data initiatives should be underpinned by active stakeholder engagement (whether within or outside the government) during design and rollout and tested by target users. There also ought to be strong mechanisms in place to safeguard personal information of users to create a virtuous cycle of trust and use of data.
- **A combination of offline and online components.** Offline tools and processes, such as communicating data via analog methods or offering in-person alternatives to online applications, continue to be vital for tech-driven applications to be adopted and succeed.

Source: This box draws heavily on Kumar, Sahu, Lal Das, and Jayakumar 2023.

2. Institutional arrangements

As Mozambique builds out its data governance framework, it will be important to clarify institutional roles and establish mechanisms for coordination and adequate financing. International experience also shows the need for champions at various levels of government to advance the data agenda. This means

[135] Eneanya 2021.

that in addition to top-down leadership (e.g., from the proposed Steering Committee), successful implementation of the data agenda will also partly depend on identifying champions across the institutional landscape who can drive new initiatives.

Specific recommendations for consideration are as follows:

a. **Ensure that key data institutions (INTIC, INAGE, INE, and others) are included in the proposed Inter-Ministerial Digital Transformation Steering Committee and the working groups it oversees.** The Steering Committee should include representation from MCTD, INTIC, INAGE, INE, MoF; related agencies such as ADE and CEDSIF; and potentially other public entities, with INTIC playing the role of convenor and coordinator on overall data strategy and governance. As international experience has shown that proximity to the highest level of government facilitates coordination of the data/digital agenda, the Steering Committee would ideally be situated under the Prime Minister's Office. The Steering Committee could oversee various technical working groups, notably on the implementation of a data strategy, or on digital interoperability. The latter has already been established, and has representation from multiple government entities, including INAGE, DNIC, DNRN and the Revenue Authority (AT). The composition of this technical group could be expanded to other key stakeholders on the data side.

b. **Clarify institutional roles and responsibilities for government entities.** Several key roles that have not yet been clearly defined in Mozambique include:

   1) **Host of government data catalog and data standards.** INAGE is the technical implementer of the interoperability framework and hence should serve as the central facilitator of data exchange across government. These functions are closely intertwined with INAGE's envisioned role as the platform for provision of digital services. In conjunction with all of this, INAGE could take on the task of maintaining an up-to-date government data catalog and hosting all data standards to be used across government, together with an eventual services meta register. INE could provide guidance on data management and quality assurance across the data life cycle and advise on data definitions and nomenclature.

   2) **Data steward.** The digital era has led to a proliferation of data that can complement traditional government statistics.[136] One significant development is the explosion of administrative data resulting from the digitalization of government systems and processes. There is also the rise of big data, citizen-generated data, geospatial data, and other forms of data. This prompts the need for a new "data stewardship" role within government. While national statistical offices (NSOs) have traditionally focused solely on official statistics, an institution acting as a data steward looks at all potential sources of data and considers 1) whether a dataset meets minimum adequacy standards for use; 2) how a dataset can help fill gaps in official statistics; and 3) how to make different forms of data accessible to users within and outside government. In Mozambique, INE could potentially expand its scope of work to assume the role of data steward, proactively

[136] PARIS21 and Mo Ibrahim Foundation 2021.

looking for ways to leverage administrative and non-government sources of data to meet the government's needs for analytics.

3) **Data leaders at the sectoral level.** Leveraging data effectively for development is first and foremost the responsibility of line ministries conducting data analysis, program management, and policy design. As a medium- to long-term goal, the government should aim to resource every line ministry with a data focal point (such as the Chief Information Officer) who would oversee the ministry's implementation of the data agenda, compliance with the data governance framework, and coordination with central institutions such as INAGE and INE. Line ministries should also be the business owners of data standards that fall under their areas of work and should ensure that these data standards (hosted by INAGE) are updated as needed, in collaborate with INE.

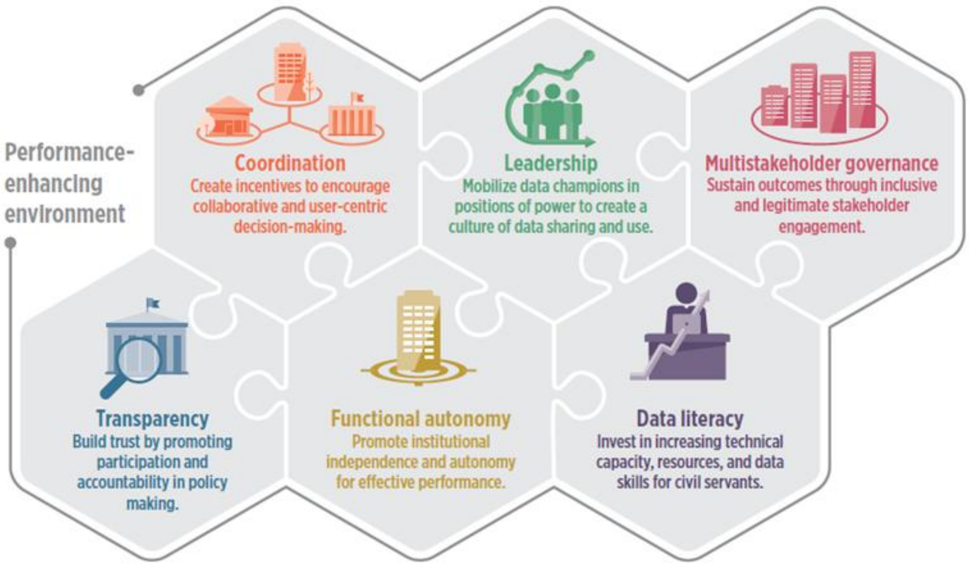
c. **Capacity building and adequate funding will be necessary to enable existing institutions to take on these additional responsibilities.** It will be important to ensure close coordination between MoF and key data institutions including MCTD, INAGE, INE, and others to secure adequate funding for a sustainable approach to the data agenda.

Lessons from International Experience

Certain institutional characteristics are particularly important for data governance institutions to fulfill their roles. These characteristics are highlighted in Figure 4 and underscore the kinds of practices that help to foster a healthy data ecosystem. Themes include the need for a collaborative, multi-stakeholder approach motivated by the needs of users and based on transparency and trust. The Government of Uruguay provides a good example of strategy-driven institutional arrangements for data governance. Uruguay's Agency for Electronic Government and Information and Knowledge Society (Agesic) was established in 2007 to spearhead the country's e-government reforms. Because Agesic is situated under the Office of the President, it has been empowered via high-level leadership to drive a multi-stakeholder digital transformation of government, ensuring that platforms, systems, policies, laws, standards, and institutions are coordinated and inclusive.[137]

[137] Example drawn from World Bank 2021a.

Figure 4. Features of high-functioning data governance institutions



Source: World Bank 2021a.

The institutional setup for data governance is unique in every country and reflects preexisting institutions, political economy, and practical considerations, but some general observations are possible:

- While 53 percent of high-income countries have a dedicated data or data governance entity, no low-income countries do (as of 2021), instead embedding these functions in other existing institutions.[138]
- Some countries have a centralized approach to data governance, with one entity taking the lead. Other countries divide data governance functions among a network of ministries and agencies. In low-capacity settings a more centralized approach is generally preferred.
- Situating the data authority at the highest level of government makes coordination easier, especially during the start-up phase, but institutional arrangements have evolved in countries as the digital and data landscape have rapidly changed. For example, the Malaysian Administrative Modernisation and Management Planning Unit (MAMPU) was initially located under the Prime Minister's office, but later transitioned to the newly-formed Digital Ministry.
- Often an IT/Digital Ministry performs the bulk of data governance functions. Ethiopia relies on the Ministry of Innovation and Technology; India, the Data Governance Division of the Ministry of Electronics and Information Technology; Jordan, the Ministry of Digital Economy and Entrepreneurship; and Rwanda, the Rwanda Information Society Authority "RISA" under the Ministry of ICT and Innovation.[139] In some countries, the NSO plays this role, such as in Estonia, where Statistics Estonia is responsible for coordinating data governance, managing the government's data classification system, and providing data sharing services.

[138] World Bank 2021a.
[139] 2022 GTMI dataset, available at https://www.worldbank.org/en/programs/govtech/gtmi.

- Data intermediaries and non-government institutions are essential components of a healthy data ecosystem, as they help promote access to data and package data so that it is more useful for non-specialists.[140]
- Institutions that establish and enforce data governance laws should be setup as independent agencies to minimize political interference and to foster public trust.[141]
- Implementation gaps—that is, the distance between laws and processes on paper and what is done in practice—are often significant when it comes to data governance. Closing the implementation gap depends on many factors that have already been discussed, as well as creating positive incentives for compliance and a culture of performance improvement. Investing in change management is essential.[142]

A challenge many countries face is that the entity charged with implementing the data/digital agenda, often a line ministry, lacks the leverage to drive the adoption of standards and common approaches across the government. One way to address this is through the creation of a high-level committee to provide cross-sectoral leadership on the data and/or digital agenda. For example, Tunisia established a Special Inter-Ministerial Committee on Digitalization to ensure coordination around the country's digital transformation strategy.[143]

3. Policy and legal framework for data governance

The establishment of a robust policy and legal framework is crucial for building trust in the data ecosystem and enabling the secure and ethical use of data. As discussed earlier, the policy and legal framework for data governance encompasses a range of issues. The recommendations here begin with data protection because the government has already identified it as a priority, the findings of the diagnostic support this analysis, and work to put data protections in place is underway with World Bank support. The focus on data protection is not meant to imply that other areas are unimportant, but rather that this is a good next step in building out the policy/legal framework. Moreover, as new policies/regulations are defined and adopted, it will be important to ensure their alignment with existing elements of the data governance framework. This includes new regulatory instruments that are currently under development, namely the proposals for Regulation on the Development, Contracting and Operation of Cloud Computing Platforms and Regulation on the Construction and Operation of Data Centers.

Recommendations for consideration include:

a. **Adopt a data protection law aligned with best international practices, while being sensitive to the local context.** Working to ensure data protection will help Mozambique align with regional standards and foster a safer data environment, laying the groundwork for data to play a greater role in strengthening governance and accelerating development, including via increased cross-border data flows. To the extent that government entities have existing rules or regulations related to data protection, these should be updated to align with the new law, which would take

[140] World Bank 2021a.
[141] Ibid.
[142] Ibid.
[143] Davenport, Kumagai, and Sylla, et al., 2024.

precedence in case of any discrepancy. A legal review of the draft proposed law will be undertaken as part of the World Bank's MDAP Project.

b. **Create and operationalize the National Data Protection Authority, giving it adequate institutional mandate and resources to be sustainable and independent.** The draft law envisions an independent National Data Protection Authority (ANPD) that will oversee compliance, issue guidelines, conduct audits, and impose sanctions. Organizations, of a certain size, will be required to appoint Data Protection Officers (DPOs) to ensure adherence to legal requirements, manage data subject relations, and maintain confidentiality. The World Bank's MDAP project includes support for operationalizing the legislation with the creation of a fully functional Data Protection Office. One question to clarify is where the office would sit institutionally. It will be key to ensure that the ANPD has sufficient teeth and implementation capacity (Box 5). There is also the question of whether the office needs to be a stand-alone entity or whether it could be embedded in an existing agency (very few low-income countries have agencies dedicated solely to data protection). Also, with a view to enabling cross-border data flows, the ANPD could join the African Network of Personal Data Protection Authorities (NADPA/RAPDP), which currently has 18 country members and aims to be a platform for exchange and coordination.[144]

c. **Revisit provisions of the Licensing Regulation in light of broader data governance objectives.** As mentioned in the findings section, there is a risk that the licensing requirement and fees could have unintended negative consequences in terms of building trust and dynamism in the online environment. It is recommended that INTIC undertake regulatory impact analysis to gather empirical evidence on potential market impacts, shift from a licensing to a registration-only approach, and create a tailored regulatory framework for online platforms and content to support the principles of a free, open, peaceful, and secure cyberspace. It will be important to ensure that goals are feasible, accounting for regulatory capacity constraints, and to identify alternative policy interventions to achieve separate economic objectives (for example, considering the OECD two-pillar approach if the policy objective is related to taxation).

d. **Strengthen implementation/enforcement of existing laws and lay the groundwork for developing wider-ranging data policies covering the entire data lifecycle, including coverage of aspects and concerns related to the use of data by AI systems and platforms.** For instance, the Microdata Dissemination and Access Policy envisions a collaborative relationship between data providers and users, ensuring that statistical outputs meet practical needs and expectations. The policy identifies key user segments, including government, Parliament, the private sector, international agencies, academia, persons with disabilities, and the general public, ensuring tailored data access and usability. Its implementation strategy focuses on provincial dissemination, creating new products, and both inter-institutional and public dissemination, and emphasizes using advanced technologies to enhance data accessibility and diversify data formats. It would be useful to take stock of the extent to which implementation has progressed and what is needed to support these efforts. Also, as mentioned in the Main Findings section, implementation of the Right to Information legislation could be strengthened to ensure more consistent public access to government information. The government could include a medium- to long-term roadmap to fill in gaps in the legal framework as part of its data strategy (see

[144] https://www.rapdp.org/

recommendation #1). The rapid evolution and adoption of AI technologies globally also raises many issues that will need to be considered not only in the National AI Strategy, but across the data governance framework.

Box 5. Empowering data protection authorities

Data protection authorities (DPAs), like other government agencies, can be susceptible to internal and external influences that either erode their authority over time or delegitimize them in the eyes of their stakeholders. The European Union has thus laid down the following requirements[145] from member countries to ensure the independence of DPAs:

- **Freedom from external influence on their leaders**, whether direct or indirect; for example, freedom from governments' or other third parties' influence;
- **The opportunity to ensure the integrity and impartiality of their leadership**, meaning that DPA members should not engage in any actions or occupations that are incompatible with their duties;
- **The financial, human and technical resources**, premises and infrastructure necessary for the effective performance of their tasks and exercising of their powers;
- **Their own staff** chosen by and under the sole supervision of the DPA member(s);
- **Financial control** that does not affect their independence;
- **Separate annual public budgets**, which may be part of Member States' budgets; and
- **Transparency** in the nomination of DPA members.

Lessons from International Experience

The African Union Data Policy Framework 2022 emphasizes the importance of data governance, protection, and the harmonization of data systems across African nations.[146] It highlights the need for cooperation, integration, and trust among member states to facilitate cross-border data flows and improve data ecosystems, and provides a comprehensive policy guide for African countries to develop and implement data governance frameworks that are context-specific yet aligned with continental best practices.

Most African countries have general data protection legislation in place, and legislation tends to be sound. According to the World Bank's *Regulating Digital Data in Africa* report, "governments that have introduced an overarching data protection law have tended to include what are emerging as common elements of good international practice in this area, such as purpose limitation, data minimization, and data subject rights, which feature in sources ranging from the OECD Principles and [the EU's] General Data Protection Regulation (GDPR)"[147], which has inspired many countries in crafting their own regulations (see Box 6). Thinking about which peers to look to for model approaches, countries should consider a variety of factors, including 1) if they want to support regional or continental harmonization or global

[145] FRA 2024.
[146] African Union 2022.
[147] World Bank 2023d.

interoperability; 2) who their main trading partners are; and 3) if they have particular policy objectives that are incompatible with approaches in specific models.

However, while the laws that are on the books in the region are strong overall, implementation has been generally slow. Countries that have adopted data protection laws have also included a requirement to create a data protection authority (DPA). For many countries it has been cost-prohibitive thus far to establish a stand-alone authority and they have instead embedded the function in an existing agency. The African Union (AU) Convention on Cyber Security and Personal Data Protection (the "Malabo Convention"), which Mozambique has ratified, includes the requirement for a data protection authority. Another issue is the cost of compliance for the private sector. For example, in Kenya, some lawyers have found that "the private sector has inadequate financial, technical, infrastructural, and human capacity, rendering compliance cost prohibitive", and "for companies already in compliance with other countries' data protection rules, it is almost impossible to separate procedures for specific customers based on their nationality or location", especially for smaller firms.[148]

Box 6. Data Protection Legislation and Implementation in Africa

A recent study found that although data protection may be costly for the private sector to implement, it is relatively affordable for governments to enforce. Following an analysis of the costs of compliance and enforcement, experts provided the following suggestions for developing countries:

Provide clear guidelines for compliance, ensuring a clear distinction between binding law and suggestions for best practices.

Avoid data localization requirements (except where justified by national security or data sovereignty), as these provide no additional security and raise costs for local firms.

Seek mutual recognition of data protection laws to reduce firms' costs of complying with several different rules when entering foreign markets.

Provide flexibilities within the law, such as exceptions for certain requirements for Micro, Small and Medium Enterprises (MSMEs), or ex post facto liability for certain infringements.

Provide alternative models for cross-border data transfers, such as self-certification, which allow transfers to countries without an adequacy decision from the European Union subject to certain conditions.

Source: Excerpt from World Bank 2023d, citing Chander, et al., 2021.

4. Procedures, practices and standards for data management

The lack of standardized procedures and controls for data management undermines data quality, security, usefulness and availability in Mozambique. Ongoing work financed by the World Bank aims to strengthen data management practices and lay the foundations for interoperability. The MDAP Project includes support for a review of data management practices and "development of a data governance framework and implementation roadmap, including the definition of shared principles, norms and standards for data governance, data policies and strategies and a shared approach to data management and hosting." At the same time, the EDGE project is supporting operationalization of some elements of interoperability. The recommendations below are envisioned as part of the process of creating this comprehensive, formal data governance and interoperability framework. Developing and enforcing data

[148] World Bank 2023d.

standards and management practices across government agencies could begin with priority sectors and data categories as identified through consultations with stakeholders (i.e., do not tackle everything at once). Note that infrastructure is also a critical enabler of effective data management practices, but outside the scope of this report.

Recommendations for consideration, that could be implemented in conjunction with ongoing work, include the following:

a. **Prepare and implement a Data Classification Policy defining the processes and standards that comprise a national security classification system.** The Data Classification Policy would define categories of data from a security perspective (for example, Top Secret, Secret, Confidential, Official, Public) and provide clear guidelines regarding how to classify, handle, and protect sensitive government data. The Data Classification Policy would provide the basis to decide which data can be made public, which data can be stored in the cloud, and which data needs to remain in in-country data centers at different levels of security. Developing such a policy and associated regulations is a priority because of its implications for a number of laws, policies, regulations, and strategies currently under development, including the Data Protection Law, the laws on cybersecurity and cybercrime, the National Cybersecurity Policy and Strategy, the National AI Strategy, and the Cloud Computing Regulation.

b. **Conduct an inventory of the data that currently exists within government to identify areas of complementarity, duplication, and overlap.** This exercise could be undertaken by INAGE (focusing on administrative data) in conjunction with INE (as data steward, with a broader view of data) and external support under the World Bank's EDGE project, and could focus initially on several high-priority sectors, including one or two that are more advanced in terms of the use of data (e.g., health) and a few where use of existing data is relatively limited and increased use of it could be impactful (e.g., education, water).

c. **Determine an institutional owner for each category of data to ensure there is a "single source of truth" and shared nomenclature.** This would mean identifying one data-producing institution (usually a line ministry) to "own" each category of information, and that agency's data would become the reference point for any other government entity that needs the information. This institution would also be the "owner" of any associated data standards and would provide corresponding APIs to other entities for their use.

d. **As an initial pilot of this multi-stakeholder approach to interoperability, the government could consider harmonizing the use of geographical identifiers across the government, building on the work of ADE and IDEMOC.** Agreeing on a common reference for geographic nomenclature to be adopted by all ministries and in all online government forms and systems would make it easier for agencies and IT systems to talk to each other and would facilitate data analysis and coordination of policies across government (see Box 7). Significant work on coordination of geospatial information is already underway. Thus far, ADE has developed strategic partnerships with government agencies such as the former Ministry of Finance and Economy (MEF), Ministry of Health (MISAU), Bank of Mozambique, and INE. From building tools for geospatial data collection and hosting, to analysis and visualization, ADE has already enabled several government agencies and ministries to use geospatial data for more informed decision making. For example, the Ministry of Health used ADE's analysis for decision making during the COVID-19 pandemic, while the Metropolitan Transport Agency (AMT) utilized telecom data for a mobility study to understand the flow of people in the Maputo metropolitan area. Coordinating all of these efforts under the umbrella of the government's larger data strategy could facilitate even more effective data applications and create a basis for the initial interoperability use cases that have already been envisioned as part of the World Bank's EDGE project.[149]

Box 7. Strengthening a Multi-stakeholder Approach to Data Governance: Geospatial Names Pilot

The importance of geospatial data is increasingly recognized in policy making and government programs.[150] As the Government of Mozambique designs and operationalizes an interoperability framework for data exchange that underpins digital services, data sharing and data analytics, geospatial data could be a powerful starting point with relevance for all sectors and ministries.

Mozambique has advanced in the realm of geospatial data through a 2023 decree that created the National Infrastructure for Spatial Data in Mozambique (IDEMOC) that regulates the effective production, sharing, integration and publication of spatial data in the country.[151] Through its National Geospatial Development Agency (ADE), the government is institutionalizing spatial planning methodology in decision making processes. ADE is tasked with enhancing the spatial development planning capacity of priority national government institutions and agencies and promoting the adoption of this methodology across the public and private sectors.

Linking this existing work on geographic data harmonization to the government's centralized effort to build an interoperability framework could be a pilot initiative for the proposed Inter-Ministerial Data Steering Committee. The Steering Committee would spearhead a process around setting government-wide geographic data standards that could then be rolled out for other types of data. The multi-stakeholder approach would ensure good coordination among the key players involved, including IDEMOC/ADE, INAGE, INE, sectoral ministries such as the Ministry of Land and Environment, and many others.

The process could begin with establishing which government entity is the business owner for the data standard in question (in this case, IDEMOC/ADE for geographic names), and enshrining this mandate in the central reference data architecture hosted by INAGE. A communications strategy would be developed (under the aegis of the public sector digital transformation strategy) to raise awareness across government that (with provisions for a transition period) an Application Programming Interface (API) provided by IDEMOC/ADE should be used as the reference for geographic names in all government applications, systems and analytics.

e. **Building on the proposed geospatial names pilot, INAGE could proceed with development of a common data exchange infrastructure (potentially leveraging the DPI model), standard practices on data storage and sharing, and data registers to standardize data definitions and enable interoperability**. The long-term vision would include a "Registry of Registers", or meta catalog, to serve as a centralized directory of all government data, compiled and hosted by INAGE, and digital IDs as a building block to make systems interoperable and allow sharing of data safely and securely across applications (as has been demonstrated in India and elsewhere). As part of this, it would be important to establish clear standards and guidelines for ensuring data quality across government (INE could advise on this).

[149] Per the May 2024 Implementation Status Report for the World Bank EDGE project, these include request of new ID, duplicate birth certificate, driving license, business registration, Criminal Record Certificate, marriage certificate, vehicle property title, and survivor's pension certificate, which are among the most requested services.
[150] https://www.un.org/disabilities/documents/rio20_outcome_document_complete.pdf
[151] World Bank 2023e.

f. **Create a mechanism to improve coordination among key data stakeholders within government in terms of data practices, infrastructure, and technology.** The government could expand the membership of the existing technical working group on interoperability, and this group could share good practices across government and look for efficiencies. For example, INE plans to transition to SDMX for data exchange[152] and create a data warehouse to enhance data accessibility and utilization, and the Bank of Mozambique also intends to adopt a data warehouse as part of its internal database integration project. This could be a good opportunity to exchange experiences and share lessons learned with the wider government. Practices and procedures should also be developed to implement the government's commitment to strengthening gender statistics and other data priorities.

g. **Support expanded and more streamlined public access to data, taking an approach informed by what current and potential data users need.** Improved coordination could help to ensure that datasets already available via the government's open data portal are up to date (which is not always the case now), as would a review of current workflows to ensure that processes are in place to publish data online when it becomes available. It could also be helpful to streamline the process for data requests from the public, such as by providing a web form. More broadly, understanding the demand side for data, and the needs of current and potential data users, should drive decisions about which datasets to prioritize for publication. Eventually, a goal would be to create feedback mechanisms through which government agencies can receive information about how their data are used and how they could be improved.[153] The World Bank's Systematic Country Diagnostic recommends that the two highest priorities for increased data transparency are 1) reporting by Mozambique's state-owned National Hydrocarbon Company (Empresa Nacional de Hidrocarbonetos (ENH) given the significance of liquified natural gas in Mozambique; and 2) the budgetary process, where progress could be monitored via the country's score on the Open Budget Index.

h. **Develop guidelines for measuring the impact and quality of data-driven services (i.e., digital public services), and publish the data online to increase transparency and accountability.** Administrative data generated continuously by digital services can help government to understand who is being served, who lacks access, and how services can be improved, leading to more inclusive and user-centric service delivery. As envisioned by the World Bank's EDGE project, data on services provided (e.g., service volume, delays) should be published online. It can also be "complemented by data collected through a proactive citizen engagement mechanism, whereby citizens are contacted through a variety of means (for example, SMS, voice) to report on issues and to provide feedback on the quality of services just received."[154] Once this data is available to government and the public, the final step to realize the value of the data is to undertake analytics and use the findings to make corrections and improvements.

i. **Develop and implement standardized controls and procedures to ensure data protection and security across government agencies:**

[152] With support from the African Development Bank.
[153] For example, the National Institute of Statistics of Rwanda conducts regular satisfaction surveys of users of official statistics. Source: Ndemo, Ndung'u, Odhiambo, and Shimeles 2023.
[154] World Bank 2021c.

1) **Enhanced data security measures.** Mozambique could strengthen data security measures by adopting international best practices including encryption, access controls, and regular security audits to prevent unauthorized access and data breaches.

2) **Detailed documentation and record-keeping.** Mozambique could adopt comprehensive documentation and record-keeping practices to ensure transparency and accountability. This includes maintaining detailed records of data processing activities, consent forms, and data breach incidents.

3) **Detailed data breach notification requirements.** It would be helpful for Mozambique to establish clear protocols for data breach notifications, including mandatory reporting to a supervisory authority and affected individuals. This will improve response times and mitigate the impact of breaches.

4) **Comprehensive compliance and audit processes.** It would benefit Mozambique to introduce regular compliance audits and assessments to ensure adherence to data protection standards. This can help identify gaps and areas for improvement in data security practices.

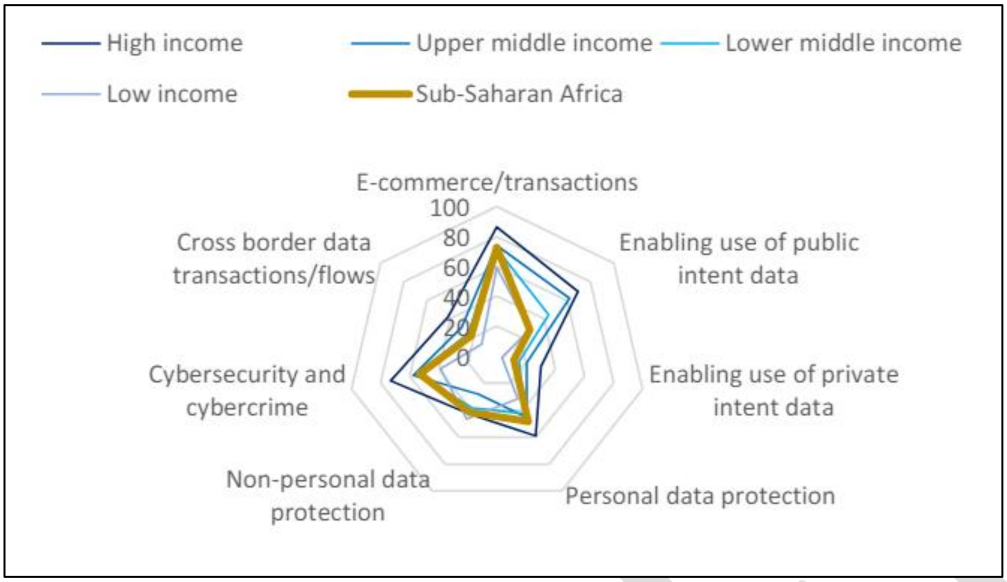
Lessons from International Experience

Looking at the region overall, there are some common challenges facing governments. Figure 5 shows the average performance of countries in Sub-Saharan Africa on key elements of data governance relative to income-based country groupings, based on the World Bank's Global Data Regulation Diagnostic. This evidence suggests that Sub-Saharan Africa particularly lags other regions in enabling use of public intent data, which includes practices such as the use of technical standards for system interoperability and common data definitions across the government, as well as open data policies and right to information policies that are implemented effectively.[155] Following principles and practices of interoperability (Box 8) can enable governments to get more from the investments they make in data.

However, there are examples of data excellence in the region. For instance, Kenya (see Box 9) offers an interesting case study of a coordinated national approach to maximizing the use of data for development. Part of its success lies in the data management practices it has adopted, such as the behind-the-scenes coordination and standards that enable the unified digital services portal, and investment in interoperability via the Health Information Systems Interoperability Framework. Kenya's efforts to enable healthcare providers and patients to access the same data, for example, is a significant step toward user-centric service delivery.

[155] World Bank 2023d.

Figure 5. Average scores on different data governance dimensions by income group/region



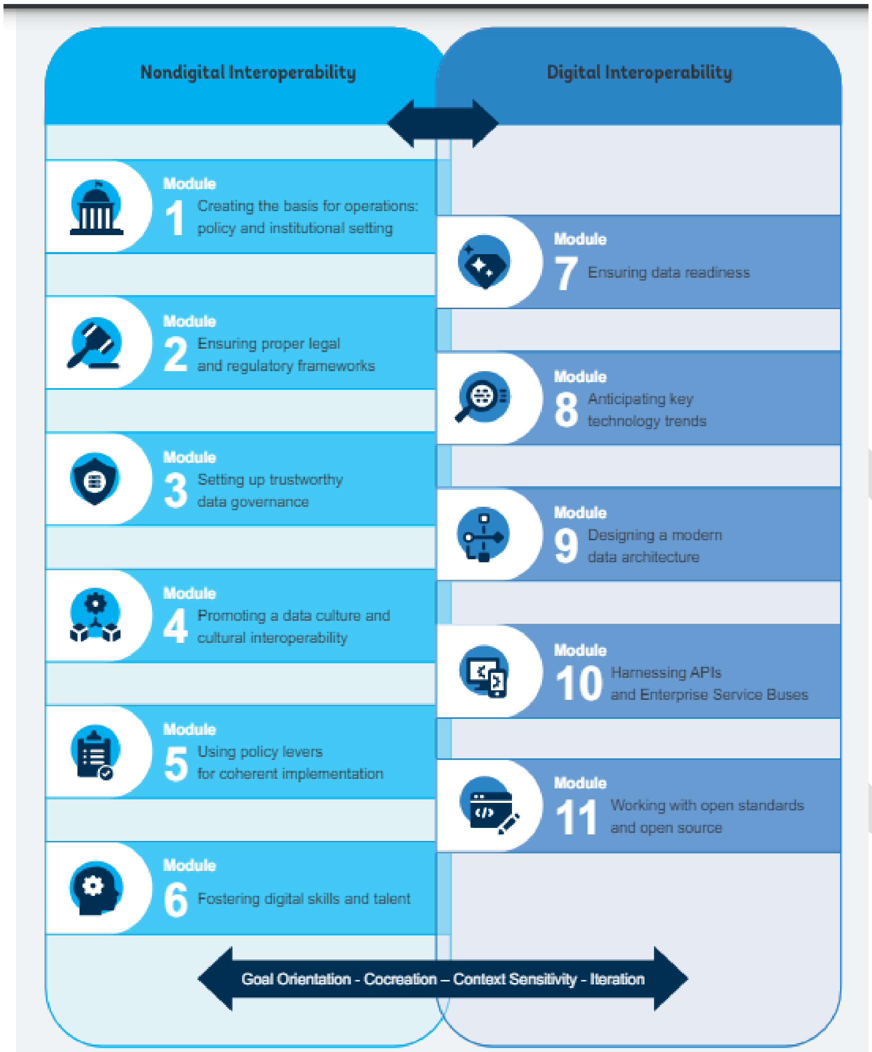
Source: Reproduced from World Bank 2023d.

Box 8. Creating interoperability in the public sector

Interoperability in government systems refers to "connections between ministries, departments, agencies, sectors, government levels and countries through data, information systems, legal agreements, organizational processes, and shared values and customs… As such, interoperability is considered as a multilayered concept consisting of both nondigital and digital elements, namely legal, organizational, cultural, technical, and semantic layers as well as their overall governance."

Interoperability allows data from multiple government sources to be easily shared and combined so that data gaps are filled, duplication of effort in data collection and processing is minimized, and new insights can be revealed from analyzing disparate datasets. It also reduces administrative burdens on citizens since it cuts down on duplicate data requests from the government when they access services (the "once-only" principle means that an individual should only be asked for a given piece of information one time).

By allowing entities across the government to share datasets that can "talk" with each other because they are based on the same definitions and standards, interoperability helps governments move from siloed approaches to data collection to a more strategic, government-wide use of a data for policy making, service delivery and program monitoring. It also makes it easier for non-government actors such as firms or researchers to repurpose public data. The side-by-side mapping of digital and non-digital aspects of data interoperability (below) is a good illustration of the importance of considering the data and the digital transformation agendas jointly—in the digital age, one cannot exist without the other.

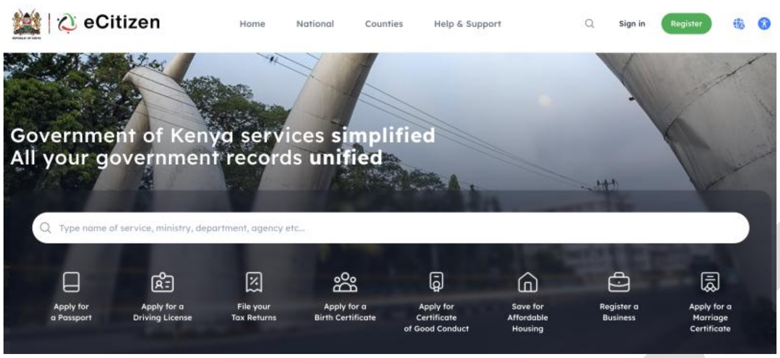


Source: World Bank 2022c.

Box 9. How Kenya leverages data for development

Kenya's approach to data-driven development is notable for its balance between top-down leadership (e.g., a National Digital Masterplan and its flagship programs) and bottom-up initiatives. At the top, the National Digital Masterplan (2022-2032) provides a blueprint for the government's foundational digital infrastructure, digital services, skill development, and regulatory environment.[156] One of its flagship programs is an e-Citizen Portal[157], a one-stop-shop making 5,000 government services across 25+ ministries and government agencies readily accessible. By offering services such as driver's license and passport applications, vital registration, and business registration, the portal aims to streamline public administration and minimize corruption by reducing physical contact between citizens and intermediaries.

[156] Ministry of ICT, Innovation and Youth Affairs (Kenya) 2022.
[157] https://accounts.ecitizen.go.ke/en#main-content



Taking the example of the health sector, the e-citizen portal hosts 330+ programs and e-services, including key features like the National Health Record, Electronic Claims Management System, and Patient Care system that empower Kenyans to manage their health records and engage in interactive sessions with instant feedback. By eliminating intermediaries, "this initiative ensures citizens have full control over their health records, facilitating seamless collaboration between healthcare providers and the public".[158] The portal also empowers 100,000 Community Health Promoters with access to Ministry of Health data, and hosts an e-procurement system for health suppliers and job boards for medical professionals, further revolutionizing healthcare delivery. Underpinning this digital offering is the Kenya Health Information Systems Interoperability Framework (KHISIF), which aligns with the Ministry of Health's strategy of "providing patient-centric, integrated health services".[159]

At the local level, counties have Data Desks[160] that collect, analyze and publish data from diverse sources for the benefit of the government and citizens. Kenya also uses citizen-generated data (CGD) to fill gaps in official SDGs reporting.[161] The Kenya National Bureau of Statistics (KNBS) validates CGD using the Kenya Statistical Quality Assurance Framework (KeSQAF), a tool for assessing and monitoring the quality of data produced in the National Statistical System (NSS) and beyond.[162] For example, the Bureau is using CGD to monitor the use of contraceptives in Kenya.[163] Kenya's subnational Data Desks and CGD initiatives strengthen participatory processes and citizen engagement while enhancing data availability.

5. Capacity building

To address skills gaps, the government could implement training programs and capacity-building initiatives aimed at upskilling civil servants in data management and analytics. These programs would focus on fostering a data-literate culture within the public sector, enabling civil servants to effectively utilize data in their daily operations. It is important also for Mozambique to develop and implement continuous training programs for employees at all levels on the country's data protection laws, policies, and procedures so that all civil servants can follow them. This will foster a culture of compliance and

[158] Ministry of Health (Kenya) 2023.
[159] https://www.data4sdgs.org/index.php/resources/kenya-health-information-systems-interoperability-framework
[160] Kags and Davenport 2019.
[161] KNBS/P4R/PARIS21 2022.
[162] https://www.knbs.or.ke/citizen-generated-data/#:~:text=The%20Kenya%20National%20Bureau%20of,System%20(NSS)%20and%20beyond
[163] https://www.knbs.or.ke/wp-content/uploads/2023/09/Citizen-Generated-Data-Perfomance-Monitoring-for-Action-Kenya-National-Phase-2-Panel-Results.pdf

awareness within organizations. Collaboration with international organizations and academic institutions can provide valuable models and resources for developing effective training programs. By investing in capacity building, Mozambique can ensure that its civil service is well-equipped to manage and utilize data effectively.

Recommendations for consideration include the following:

a. **Conduct an inventory of data skills gaps in key roles within government institutions that are important to the data governance framework.** There is a need for a plan supported by adequate resources to identify which skills/tools are most needed where, and how to develop them either through upskilling of current staff, recruitment, or outsourcing/partnerships.

b. **Think beyond technology and focus on building competencies such as digital business process engineering, digital business and operational models, digital market structures, and digital consumer and citizen rights**. These skills are crucial for government officials to conceptualize and roll out digital public services, regulate digital businesses, and safeguard the interests of citizens and consumers during digital transactions.

c. **Establish general data training programs for the civil service at different career stages and invest in continuous upskilling.** Under the EDGE project, the World Bank is supporting advanced digital skills training for civil servants. INAGE has signed an MOU with UEM and is also in discussion with the Ministry of State Administration to see how to incorporate digital skills training into career management for civil servants. The project will also finance an assessment of digital skills gaps in government and on that basis will fund pilot training programs in cooperation with local and international universities, aiming to design a sustainable model. It is recommended that data literacy and analytics skills could be embedded in this broader civil service training program. Efforts to enhance data skills within the civil service should involve collaborations with academic institutions to develop tailored programs addressing the specific needs of government officials.

d. **Establish specific training courses on aspects of data governance for select staff of key agencies who need to be well versed.** This could start with training of staff in the new Data Protection Authority but eventually should include all regulatory agencies that have oversight on data related issues (e.g. consumer protection or competition authority).

e. **Talent management: fellowships/rotations for data (e.g., two years in different agencies).** Government jobs, around the world, have a reputation for blunting technical skills over time. One way to address it is to provide motivated staff with opportunities to move across government agencies if they have cross-cutting skills like data governance. The Free Agent model[164] introduced by the Government of Canada is a potential model.

f. **Cross-pollinate good practices across the government and encourage learning by doing.** Knowledge management and fostering a culture of knowledge sharing among institutions will help the government build on its successes.

Lessons from International Experience

Prior analysis of digital capacity development in Mozambique and other African countries has helped identify several design and implementation strategies that may increase program effectiveness. These include (some drawn from Rwanda's Digital Ambassadors Program):

[164] See https://wiki.gccollab.ca/Canada%27s_Free_Agents/FAQ.

- Take a systemic view incorporating the formal educational sector and programs offered by other government, private, and civil society actors
- Develop linkages between and across different programs
- Incorporate local languages, learning techniques, and partners into training programs
- One size does not fit all so develop variegated programs depending on the context
- Emphasize coordinated trainings over time rather than ad-hoc capacity building initiatives; trainings/courses should be defined as part of a programmatic plan outlining a clear progression in skills development such that each installment builds on what came before, with focus areas based on an inventory of skills gaps—the goal being to build and institutionalize skills in a sustainable way
- Digital technologies change rapidly so be prepared to develop and deliver constant refresher material
- In the same vein, take advantage of emerging technologies to improve the quality of training
- Be sensitive to local cultural preferences and norms that might create participation barriers
- Work with the community and its representatives

Bringing external talent into government can be another important way to address skill gaps. New staffing models allow successful entrepreneurs and successful technologists to contribute to public sector initiatives. Examples that the Government of Mozambique might consider include:

- **Digital fellowships with government.** Data or digital fellows are data experts hired from the private sector or academia for short- to mid-term engagements (typically 12-24 months) to work on mission-critical problems while embedded within specific government agencies. The fellows contribute as advisors or hands-on technical specialists, and bring external perspective and skills to government programs that otherwise cannot afford to hire such specialists. The Presidential Innovation Fellowship program[165], started by the US federal government in 2012, is a model for such programs.

- **Partnerships with civil society organizations to serve as conduits for talent.** Many governments work with civic-minded organizations that are deeply networked with digital professionals to identify and attract talent. The Government of Canada for instance worked closely with Code for Canada to secure placements for an incoming cohort of fellows.[166]

- **Establish an on-demand pipeline of external talent for project hiring.** Government processes and vetting requirements can make it difficult for government agencies to access external talent. The GC Talent Cloud[167] in Canada is an innovative response to the problem—it provides a curated list of vetted talent that any government agency can tap into for project-based work.

4. Next steps

Implementing the recommendations in this document is no easy task and requires coordination across diverse stakeholders. Strengthening the data governance framework will require dedicated financing and incentives to produce, protect, and share data. Implementing the recommendations of this report will further require the good judgement of partners in Mozambique, collaboration with key government and

[165] See https://presidentialinnovationfellows.gov/apply/track-ai-data-and-analytics/.
[166] Aghdaee, Kou and Craft, undated.
[167] See https://www.oecd.org/gov/innovative-government/Canada-case-study-UAE-report-2018.pdf.

non-government stakeholders, and support from the World Bank and AfDB through ongoing initiatives. While this report draws heavily on the perspectives of government and international organizations, engaging a broader range of stakeholders to understand their challenges and needs is critical. Civil society organizations can support data collection from marginalized and hard-to-reach communities, help build trust and accountability between citizens, and promote data literacy. Finally, the private sector, as both a user and producer of data, can drive innovation in data applications (i.e., data-driven services), generate and share data for more informed policymaking, and invest in critical infrastructure (i.e., data storage centers, cloud services, connectivity) where government capacity is limited.

Strategies will be needed to address central challenges and mitigate risks. The current institutional and regulatory landscape is fragmented, with agencies in the Mozambican government at varying levels of capacity and political empowerment. As such, progress toward a unified national data agenda may be non-linear or require greater capacity building than expected. This makes political momentum and strong leadership even more important. Aligning the data agenda with areas that already have strong political commitment, such as the digital transformation agenda, is one strategy to mitigate this risk.

Looking at the recommendations outlined in this report overall, immediate next steps involve generating buy-in among government leaders, specifically in assigning individuals to take on the responsibility of developing a strategic approach to the data for development agenda. This process needs to be closely coordinated with ongoing initiatives at the regulatory and technical level, such as drafting the Data Protection Law and designing the interoperability framework. Ultimately, Mozambican leadership and partners on the ground are best positioned to judge and apply the most relevant aspects of these recommendations, especially in light of an evolving political situation since the October 2024 general elections.

For a country to take advantage of data-driven development, it must develop strong digital data stocks. Mozambique is still a relatively data-poor country with few data assets. Core economic sectors such as natural resources and agriculture are still mostly not digitalized and the same is true for most government services. The digitalization of these sectors will create new data stocks that the data governance recommendations in this report can help turn into development opportunities and tools.

Future reports should aim to corroborate the hypotheses of this rapid data diagnostic and explore additional research areas. Cybersecurity policy and data infrastructure are also foundational for a functioning data ecosystem. Stronger data governance also means greater capacity for Mozambique to capitalize on opportunities offered by AI and other emerging trends to lower the cost of finding and using relevant data for policymaking. It would also allow the private sector to deepen regional trade ties, develop new products, and attract investment. Future research on the potential for policies to accelerate progress in these areas could be useful.

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Appendices

Appendix 1. List of organizations represented in consultations[168]

Government Stakeholders

Note: A reorganization in January 2025[169] resulted in new Ministry names and portfolios. Since the consultations for this report were held in 2024, the names of the Ministries as they were at the time of the consultations is listed first, with the relevant new Ministry as of the reorganization noted in brackets.

Bank of Mozambique
Directorate of Planning, Statistics and Cooperation (DPEC, at the time part of MCTES—see below)
Energy Regulatory Authority (ARENE)
Ministry of Economy and Finance (MEF) [Ministry of Finance]
Ministry of Education and Human Development (MINEDH) [Ministry of Education and Culture]
Ministry of Health (MISAU)
Ministry of Justice, Constitutional and Religious Affairs (MJCR)
Ministry of Science, Technology and Higher Education (MCTES) [now dissolved; the technology portfolio is under the new Ministry of Communications and Digital Transformation as of Jan. 2025]
Ministry of the Sea, Inland Waters and Fisheries (MIMAIP) [Ministry of Agriculture, Environment and Fisheries]
National Company of Science and Technology Parks (ENPCT)
National Geospatial Development Agency (ADE)
National Institute of Communications of Mozambique (INCM)
National Institute of Electronic Government (INAGE)
National Institute of Information and Communication Technologies (INTIC)
National Statistics Institute (INE)
Tax Authority (AT)

Non-Government Stakeholders
Eduardo Mondlane University Informatics Center (CIUEM)
Institute of Social Communication of Southern Africa, MISA Mozambique (MISA Moçambique)
Mozambican Association of ICT Professionals and Companies (AMPETIC)
Pedagogical University of Maputo Informatics Center (CIUP)
Standard Bank

[168] List includes consultations held in person in April 2024, plus additional virtual consultations.
[169] Decreto Presidencial n.º 1/2025, available at https://archive.gazettes.africa/archive/mz/2025/mz-government-gazette-series-i-supplement-dated-2025-01-16-no-12.pdf.

Appendix 2. Diagnostic questionnaire

The diagnostic team adapted the Open Data Readiness Assessment (ODRA; Version 3.1), prepared by the World Bank's Open Government Data Working Group, to prepare a questionnaire for stakeholders in Mozambique. The ODRA assists in identifying actions that a government authority could consider prior to establishing a data program either at the sub-national level, or for an individual public agency. It consists of a rapid diagnostic of the eight dimensions considered essential for the success of a data program. These eight dimensions include: 1) Senior Leadership; 2) Policy/Legal Framework; 3) Institutional Structures, Responsibilities and Capabilities within Government; 4) Government Data Management Policies, Procedures and Data Availability; 5) Demand for Data and Data Services; 6) Civic Engagement and Capabilities for Data; 7) Funding a Data Program; and 8) Technology and Skills Infrastructure.

This diagnostic lightly adapted the ODRA tool to better fit its purpose. The diagnostic emphasized the first four pillars, as well as public sector capacity (skills, under pillar 8). The current status of data demand from non-government data users and of civic engagement (pillars 5 and 6) were combined into one pillar and briefly considered given the importance of a multi-stakeholder approach in the interest of understanding the data ecosystem more holistically. The issue of funding (pillar 7) is touched on only marginally. Adequate digital infrastructure (pillar 8) is an essential enabler for a data-driven public sector, but largely beyond the scope of this diagnostic. The team added the topic of Artificial Intelligence and Machine Learning as a pillar given its increasing prominence in global policy dialogue, though it is addressed only briefly.

Key questions for each of the dimensions are below.

SENIOR LEADERSHIP:

This section includes questions on whether there is a clear vision set and implemented by senior leadership on data-led digital transformation for the country. (Suggested to be sent to the Prime Minister's Office or answered by INTIC)

A. Has the government articulated a vision/strategy for data-led digital transformation of the country (or for any priority sectors)?
    i. Please provide links to announcements or initiatives demonstrating the government's commitment to data-driven development.
    ii. Please include references to any roadmaps or performance targets included in the government's vision of digital transformation.
B. Is there a government agency/official responsible for setting national data vision, strategy, and related implementation activities in the country, and coordinating them with provincial or other sub-national initiatives? If yes, which one(s)?
C. To what extent has political leadership/government engaged in the following? For each, please provide links, references, or supporting documents to activities or programs the government has initiated:
    i. Investment in data skills and data infrastructure in the country (including broadband, data/computing platforms that provide hardware and software services to businesses and CSOs, apart from government agencies, data storage etc.)
    ii. Regulatory environment for the collection, sharing, and use/reuse of data (online, through IoT devices including sensors and drones, and for tools that utilize AI)
    iii. Promoting data innovation and entrepreneurship in the country
    iv. Raising awareness about Data among government agencies or the public
    v. Development of digital literacy and capacity building in the country, including those of government officials
    vi. Measuring the effectiveness of data ecosystem and impact of data initiatives
    vii. Socialization of Data initiatives including data platforms, services and tools related to data management and governance (e.g., through public announcements, events centering these initiatives etc.)
    viii. Mitigating harms from the misuse of data (especially in the context of marginalized and otherwise vulnerable populations) within the country and mitigating the digital divide
    ix. Security and protection of individual data

x. Cross country collaboration to address transnational challenges and cyber threats

(Additional questions)

A. To what extent do current political priorities support the different potential developmental impacts of data: transparency and accountability; economic growth; inclusion and empowerment; improving public services; and government efficiency?

B. Where is Mozambique in the political cycle? What is the scope for sustained momentum to release data/implement data diagnostic and action plan before the next elections? What are sources of challenges/potential resistances in this data initiative?

C. Which political priorities could be significantly assisted by Data?

POLICY/LEGAL FRAMEWORK[170]:

This section includes questions about how supportive/instrumental the policy and legal framework for the data-led digital transformation of the country is and where the gaps are. (Suggested to be sent to the Prime Minister's Office or answered by INTIC.)

A. Do government agencies officially sell their data and data products (e.g., reports, analyses, access to dashboards etc.)? Which one and to what extent are any such agencies dependent on revenues from sale of data for their annual budgets?

B. Does the government have exclusive arrangements (partnership agreements) with any companies with respect to sharing and using any datasets?

C. Has the government provided any guidance on the use of emerging data-driven technologies such as AI, IoT, and blockchain? Does the government provide any guidance on algorithmic-decision systems, and the use of automated or semi-automated systems by the government?

D. How does the government (or any agency or local authority) license or permit the release/use of its data? Are there policies that mandate the release of government data in specific formats?

E. What policies and systems are in place to protect against data breaches and cybercrimes?

F. Are there policies or requirements for public agencies to identify dataset custodians (e.g., GIS Officer) or to appoint a ministerial data administrator within each ministry who is tasked with coordinating data related issues?

INSTITUTIONAL STRUCTURES, RESPONSIBILITIES AND CAPABILITIES WITHIN GOVERNMENT:

This section includes questions that analyze institutional structures and capacities for data management and coordination. (Suggested to be answered by INTIC)

A. Please provide a document or organization chart that describes how data functions are organized within the government. Please identify nodal/coordinating agencies and indicate if any agencies include any officials or departments responsible for communication, information, and data management.

    i. Have any of the agencies above been assessed for relevant capabilities, project management experience, and technical skills. Please share relevant documentation if available.

    ii. Has any agency been granted the formal mandate to be the lead institution in the planning and implementation of a Data Program? Which one?

    iii. If such a mandate has been granted, does it cover relations with partners outside the government?

    iv. Are there limitations (practical, operational, technical etc.) to the execution of the mandate?

B. What inter-agency mechanisms are used to coordinate data issues (such as for technical matters)? Could you describe briefly? Would you say that they are effective? Please give concrete examples that would support your -- positive or negative -- assessment?

C. Which agencies or ministries have been the most active in the adoption of data-driven tools and platforms for service delivery, and in releasing their data? Conversely, which agencies or ministries appear most concerned about the

[170] The preliminary analysis and recommendations in this section are based on information and opinions collected from interviews undertaken and materials provided by the government and other local stakeholders during this study. This section is not based on detailed, legal due diligence and does not constitute legal advice. Accordingly, no inference should be drawn as to the completeness, adequacy, accuracy or suitability of the underlying Diagnostic, or recommendations, or any actions that might be undertaken resulting therefrom, regarding the enabling policy, legal or regulatory framework for Data in Mozambique. It is therefore recommended that, prior to undertaking any legal action to address any legal Diagnostic issue raised herein, a formal legal due diligence be performed by competent, locally qualified legal counsel, preferably assisted by international legal experts with relevant experience and knowledge of these areas.

release of data, and what is the basis of their concern? How can they be handled procedurally, and how can their concerns be addressed?

D. Which government agencies or ministries have the strongest data skill base?
   i. What part do data and/or ICT skills play in deciding civil service grades and promotions?
   ii. To what extent do government officials receive training and/or retraining on data science, data standards or data analytics?
   iii. Are there national training centers or institutes specifically established for retraining and capacity reinforcement of civil service officials in data/data science topics? Provide the most recent course catalog of such programs.
   iv. Are there academic programs in universities or training centers that civil service officials attend for retraining and capacity reinforcement on data/data science?

GOVERNMENT DATA MANAGEMENT POLICIES, PROCEDURES AND DATA AVAILABILITY:

This section includes questions about data management procedures and practices for effective Digital Governance, including norms, cultural attitudes towards data management, data ownership standards, and data archiving procedures. (Suggested to be sent to and/or discussed with relevant line ministries or answered by INTIC)

A. Has the government established a whole-of-government technical framework for the deployment of digital government and are related IT architecture and operational processes defined and established?
B. Please provide links to the practices on the management of government information. These should include references to guidelines on data security, data privacy, data archiving and preservation, and data sharing.
C. What inventories of data exist, both at whole-of-government level and by each agency responsible for data (such as the IT Ministry or Stats Office)? What standards for these apply across government as a whole? Do they include information on the formats used for production, storage, dissemination, and exchange?
D. Are processes in place to share core government data (business registry, land database, addresses) electronically across different government agencies? Do they work?
E. What government interoperability framework exists, and how is it actively used by agencies to support the development of integrated data assets and information exchange? Give examples, if any?
F. Do government agencies apply specific quality norms within their working processes and particularly in the management of the data they produce and disseminate?
G. Is there a culture of excellence regarding data management within the government? Which government structures are mostly concerned?
H. What procurement guidelines are routinely required when government acquires data from third parties? Are there any standards for ‘ownership’ (especially for data generated by contractors, and in partnership arrangements), reuse/integration, resultant IP, sharing with other government agencies?

DEMAND FOR DATA AND DATA SERVICES AND CIVIC ENGAGEMENT:

This survey section delves into the demand for government data and data services, assessing how various stakeholders, including businesses, civil society, and media, utilize government data, the types of data they seek, and the channels through which they access and engage with it, while also exploring mechanisms for civic engagement and outreach in digital and data-driven development. (Suggested to be sent to or discussed with universities, research centers, businesses, medias, NGOs, external donors)

A. What local businesses/civil society or academics/universities use data extensively? Please provide examples and use cases of the use/reuse of government data by third parties.
B. What data do they consider the government should make available? Please give the main topics/sectors demanded?
C. Please describe the demand for data infrastructure (processing, storage) from CSOs and SMEs.
D. What is the process for people to request data/report/other data products information from public agencies / government?
E. What businesses, local or branches of international firms, exist to provide value-added services to business-to-business commerce such as mobile big data, business directories, risks factors to residences or infrastructure? What government data would they like to see released? Please provide links of business or firms.
F. What programs has the government established for civic engagement and outreach related to digital/data driven development?
K. Has the government identified, and does it work with, any intermediary organizations – CSOs, media, others – to promote digital activities and build civic capacity to use data safely and responsibly? Please provide links to associated organizations and activities.

(Additional questions)

L. Which non-government communities (e.g., media, CSOs, academic, SMEs) use government data to develop products or stories to inform the public climate related or health related risks?
M. What is the level and nature of actual demand and latent demand for data from Civil Society, Development Partners and the media?
N. Has the government articulated/established processes for user-centered design of digital services? And does the design of digital services account for location, language, connectivity, gender, skills, and affordability?
O. Are there technical schools or universities with computer science, data science, statistics or quantitative programs? How many graduate with technical degrees every year?
P. How extensive is the use of data by media? What are the major local media outlets, link, and two- to three-sentence summary of recent stories (both digital and not) to provide context.
Q. How do the government/individual agencies use social media or other forms of digital engagement? What policies exist for this, if any?
R. To what extent have citizens digitally engaged with government and on what platforms? What are the main issues that generate engagement? How would citizens know if their input was considered?
S. What level of citizen-to-citizen engagement is there on key political and social issues? Is this driven by data to any extent? How would the availability of government data improve this debate?

FUNDING A DATA PROGRAM:

This section assesses funding sources and staffing allocations for data programs for innovation, public-private partnerships, and support for ICT infrastructure to facilitate data-driven initiatives. (Suggested to be sent and/or discussed with to Ministry of Finance or answered by INTIC)

A. What resources (from national budget or external support) have been earmarked for funding a data program for resilience, digital workforce, and economy? What is the source of this funding and is it contingent on additional approvals and performance targets? How/have they been utilized in the past?
B. What already exists in terms of dedicated staff for data management, both for the overall Data Program and among key agencies?
C. What funding mechanisms does the government have for data-driven innovation?
   a. What funds does the government have for development of applications or e-Services?
   b. What programs does the government have to support or promote entrepreneurship, start-ups, or development of SMEs?

c. What funding is available to build the capacity of civil society and academic community? What funding (e.g., from donors or non-government sources) is there for innovations to promote "data-enabled resilience/development"?

D. What public-private partnerships exist related to technology, that have made investments in data driven solutions?

E. What funding is available to support the necessary ICT infrastructure and ensure enough staff have the skills needed to manage a Data Program?
   a. How could common infrastructure across agencies that can be leveraged?
   b. What funding and technical skills in key agencies exist to get data supplied to a Data Portal (including curation and cleaning of data)?
   c. What Diagnostic has been made of vendor's data skills and what they might cost?

NATIONAL TECHNOLOGY AND SKILLS INFRASTRUCTURE:

This section assesses the country's digital infrastructure including internet access, smartphone usage, inclusivity efforts etc. as well as data education, computing resources, and digital skills levels. (Suggested to be sent to and/or discussed with the relevant ministry(ies) or answered by INTIC)

A. What national data/computing infrastructure has been made available to actors, within and outside the government?

B. How strong are the IT industry, developer community?
   i. What are the statistics on size of local data and ICT industry (e.g., in terms of employment, revenues, percentage of GDP etc.) and advisory/support services on data management?
   ii. To what extent do Ministries or agencies outsource IT and data management functions or services to the private sector? What kind of functions or services are typically outsourced?
   iii. Are any IT or data management services exported or imported in the international market (e.g., BPO, data centers etc.)?

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING:

This section evaluates the country's AI landscape, including skill levels, utilization of big data technologies and new opportunities in the nascent space. (Suggested to be sent to and/or discussed with the relevant line ministries)

A. Could you give examples of processes using AI technologies that have been implemented, are implanted, or are planned in the country?

B. Has the government established or plans to establish any digital/data/AI skills programs within universities/schools? Do similar programs exist for government officials?

C. How many AI experts, machine learning engineers or data scientists are employed in the government?

D. Does the government use and deploy big data (examples include cloud infrastructure, use of IoT devices, use of large-scale online data platforms, AI in services)?

E. What is the government experience with algorithmic/AI based decision-making? What processes/tools are in place to make sure algorithms are used responsibly by the government?

F. Are there any collaboration/partnership on these issues between the government and other structures from the academic world or the private sector? With other countries and or regional/international organizations?

(Additional questions)

G. What is overall level of adoption of new technologies such as Artificial intelligence/AI in the government and what does the AI skills look like?

H. How extensive is the use of Artificial Intelligence and LLM models by citizens? What are the main platforms?

Appendix 3. Pillars of the Statistical Performance Indicators

The pillars that comprise the SPI are as follows:

1. **Data Use:** The statistical system produces data that are used widely and frequently.
2. **Data Services:** A range of services connects data users to producers.
3. **Data Products:** Dialogues between data users and producers drive the design and range of statistical products.
4. **Data Sources:** The statistical system goes beyond the typical censuses and surveys to include administrative and geospatial data as well as data generated by private firms and citizens.
5. **Data Infrastructure:** A mature statistical system has well-developed hard infrastructure (legislation, governance, standards) and soft infrastructure (skills, partnerships).

For more information, see https://www.worldbank.org/en/programs/statistical-performance-indicators.